Perceived plasticity of climate-relevant behaviors and policy support among high- and lower-income individuals

Kristian S. Nielsen[1], Fabian Dablander[2,3], Ramit Debnath[4,5], Charles A. Emogor[6,7], Sakshi Ghai[8], Wencke Gwozdz[9], Ulf J. J. Hahnel[10], Wilhelm Hofmann[11,12], & Jan M. Bauer[1]*

1. Department of Management, Society and Communication, Copenhagen Business School, Frederiksberg, Denmark.
2. Institute for Biodiversity and Ecosystem Dynamics, University of Amsterdam, Amsterdam, the Netherlands.
3. Institute for Advanced Study, University of Amsterdam, Amsterdam, the Netherlands.
4. Collective Intelligence and Design Group and climaTRACES lab, University of Cambridge, Cambridge, United Kingdom.
5. Climate and Social Intelligence Lab, California Institute of Technology, Pasadena, CA, United States.
6. Department of Computer Science and Technology, University of Cambridge, Cambridge, United Kingdom.
7. Conservation Science Group, Department of Zoology, and Conservation Research Institute, University of Cambridge, Cambridge, United Kingdom
8. Department of Psychological and Behavioural Science, London School of Economics and Political Science, United Kingdom.
9. Department of Consumer Research, Communication and Food Sociology, Justus-Liebig-University Giessen, Giessen, Germany.
10. Faculty of Psychology, University of Basel, Switzerland.
11. Department of Psychology, Ruhr-University Bochum, Germany.
12. German Center for Mental Health (DZPG), partner site Bochum-Marburg

* Corresponding author

Email: ksn.msc@cbs.dk

Abstract

Widespread behavior changes are essential to mitigate climate change, including adopting improved technologies and reducing energy-intensive activities. This study examines country and income differences in the perceived plasticity of climate-relevant behaviors and their connection to climate policy support through an online survey across Denmark, India, Nigeria, and the United States (n = 4,003). In each country, 50% of participants were from the top 10% income group, a demographic systemically under-represented in behavioral research. Our findings reveal substantial heterogeneity in the perceived plasticity of investment and curtailment behaviors, with higher plasticity observed in India and Nigeria. We also identify varying relationships between income and perceptions of behavioral plasticity across different behaviors and countries. Consistent with previous studies, we find positive correlations between perceived behavioral plasticity and support for domain-matched climate policies. These results underscore the intricate relationships between income, perceptions of behavioral plasticity, and support for climate policies.

Introduction

Large-scale behavioral changes can substantially contribute to limiting climate change[1–3]. The Intergovernmental Panel on Climate Change (IPCC) distinguishes three categories of demand-reducing behavior changes[2]: (1) adopting improved end-use technologies (e.g., energy-efficient appliances, heat pumps, or electric vehicles), (2) shifting activities to less energy-intensive provisioning systems (e.g., shifting from cars to public transport), and (3) avoiding or reducing energy-intensive activities (e.g., reducing or avoiding air travel, car driving, or meat consumption). Despite the potential of behavioral changes to substantially reduce greenhouse gas (GHG) emissions[2], this potential remains mostly untapped[3,4]. Numerous factors contribute to this, including feasibility barriers common to many mitigation solutions[4–6]. These barriers include socioeconomic constraints, the characteristics and availability of climate-friendly behavioral alternatives, and often limited political and organizational prioritization of behavior change initiatives[7].

The feasibility of behavioral changes varies profoundly across behaviors, individuals, and contexts. A key concept for examining this variability is behavioral plasticity, which refers to the degree to which behaviors can be changed. Behavioral plasticity is ideally evaluated using evidence from implemented interventions or policies or high-quality field experiments to understand the malleability of behaviors across and within population groups[8–10]. However, such evidence may be unavailable or only available for some behaviors in a given context, potentially undermining the effective prioritization of behavior change initiatives.

A more practical approach is assessing perceived behavioral plasticity, which involves examining people's perceptions of the feasibility of changing their own behaviors while considering associated costs and barriers[11–13]. Perceived behavioral plasticity thus reflects indications of self-efficacy and willingness to change behavior. While perceived behavioral plasticity may imperfectly match actual plasticity, it offers suggestive evidence of which behavioral changes may be realizable, which face feasibility or willingness barriers, and how these may vary across population segments and contexts. For example, low plasticity of purchasing an electric vehicle can reflect objective feasibility barriers, such as lack of finances or limited availability of charging stations, and perceived barriers, such as range anxiety[14]. Policymakers, organizations, and other change agents can use this knowledge to inform and target their behavior change initiatives, especially when evidence of initiative effectiveness is limited.

Socioeconomic status, particularly income and wealth, shapes behavioral plasticity and people's capability and opportunity to adopt low-carbon behaviors and technologies[15–18]. For example, wealthier individuals have more financial resources to adopt low-carbon consumer technologies, a higher capacity for utilizing subsidies or tax credits (e.g., for energy-efficiency upgrades)[18,19], and a greater opportunity to outsource information-seeking and skill development related to low-carbon behaviors (e.g., finding a reliable solar-panel provider or learning to cook tasty and nutritious plant-based meals). Furthermore, wealthier neighborhoods often have more access to low-carbon behavioral alternatives, such as public transportation, active travel infrastructure, plant-based foods, and electric vehicle charging stations, which can lower adoption-related barriers[21,22]. Finally, wealthier individuals usually have substantially higher carbon footprints than the population average[23–26], thereby presenting a greater potential (and responsibility) for emission reductions through curtailing consumption.

Despite the generally higher feasibility of adopting and performing low-carbon behavior, wealthier individuals may not recognize or be willing to seize these opportunities for change[16,27,28]. For example, a recent study involving high-income participants in the United Kingdom found a prevalent reluctance against climate-relevant behavior change,

partially justified by a desire to maintain their lifestyle and privileges[27]. This finding suggests that wealthier individuals may be mainly willing to adopt low-carbon behaviors that maintain or elevate their social status[29,30], particularly those involving limited behavioral adjustment[31]. Numerous studies have found a greater willingness among wealthier individuals to adopt low-carbon technologies[16,28,29,32], such as energy-efficient appliances, which require limited behavioral adjustment to reduce GHG emissions. Conversely, these studies found a lower willingness to curtail consumption, such as reducing energy usage, often requiring substantial behavioral adjustments.

Addressing climate change requires both individual and system-level changes[2,3]. Consequently, studies increasingly explore the connections between people's past or intended behavior and their support for climate policies[33,34]. However, few studies have investigated the relationship between perceived behavioral plasticity and climate policy support[12,35]. There are good reasons to expect either a negative or positive relationship. For example, people who perceive their behavior as malleable may see less need for policies promoting behavioral change. Alternatively, people who perceive their behavior as malleable may support policies that incentivize behavior change or reduce barriers to adoption, encouraging others to modify their behavior. In the largest study to date, Kukowski et al.[12] found a positive relationship between perceived behavioral plasticity and climate policy support, particularly for behaviors with high carbon footprints. However, the study could not adequately evaluate the role of income, which likely varies across behaviors and climate policies[33], and primarily included North American and European participants.

In this Article, we investigate country and income differences in perceived behavioral plasticity of climate-relevant behaviors and its link to climate policy support. We use data from an online study conducted in four countries: Denmark, India, Nigeria, and the United States. Importantly, within each country, we recruited around 1,000 participants equally divided across two population segments: individuals whose personal income belonged to the Top 10% income bracket and those whose personal income was below the threshold for the Top 10% income bracket. The strong representation of high-income individuals is rare in survey-based and behavioral science research[36–38]. Our study therefore offers important insights into the behavioral and policy preferences of high-income individuals, whose personal carbon footprints and political influence are typically disproportionately large[18,23,24]. Representing four geographically, economically, politically, and culturally diverse countries, we also move beyond the overwhelming focus on Western countries in behavioral research[39–41], allowing us to examine potentially substantial within- and between-country heterogeneity.

Perceived behavioral plasticity across countries

Across the four countries, participants indicated their likelihood of performing five curtailment behaviors (e.g., taking fewer flights, eating less red meat) and five investment behaviors (e.g., purchasing an electric vehicle, replacing older appliances with newer energy-efficient ones) within the next 1-2 years. To measure perceived behavioral plasticity, we excluded participants who already performed these behaviors (e.g., vegetarians or those without a car) or could not perform them (e.g., not owning a house; see Methods for details). This exclusion ensured that the analyses only included participants who could realistically perform the behaviors.

We observed marked descriptive differences in perceived behavioral plasticity across behaviors and countries (Fig. 1 and Supplementary Table 1). In Denmark, a larger proportion of participants reported it as (very) unlikely that they will drive fewer kilometers (48%), take fewer flights (43%), eat less white (57%) and red meat (43%), improve home insulation (42%), install solar panels (57%), purchase an electric vehicle (48%), and move investments to climate-friendly products (46%). Conversely, more participants found it (very) likely to

use less energy for heating or cooling (45%) and replace older appliances with energy-efficient models (53%). The United States showed a similar pattern, except a larger proportion found it (very) likely to drive fewer kilometers (43%), take fewer flights (43%), and improve home insulation (47%), in addition to using less energy for heating or cooling (48%) and replacing older appliances (49%). These descriptive findings starkly contrast from Nigeria and India. The majority of participants in these countries reported high perceived plasticity, indicating that they were (very) likely to perform all behaviors (average of 57% for Nigeria and 72% for India), except for eating less white meat in Nigeria (38%). The proportional differences in perceived plasticity between countries are also reflected in the mean perceived behavioral plasticity (Supplementary Fig. 1). In Denmark and the United States, the mean perceived plasticity was relatively low for most behaviors (M = 2.80 and 2.90, respectively). In contrast, the mean perceived behavioral plasticity was considerably higher in Nigeria and India (M = 3.50 and 4.00, respectively).

There were notable country differences in the proportion of participants who reported already or never performing a behavior (Extended Data Fig. 1), with 30% of participants in Nigeria reporting that they had already installed solar panels and almost 30% of participants in India reporting that they never eat red meat. In contrast, participants from Denmark and the United States indicated much more often that they could not perform a particular behavior (Extended Data Fig. 1). For example, about 30% of participants in Denmark reported that they could not improve home insulation, while almost 40% reporting that they could not install solar panels. The proportion of participants indicating this was slightly lower in the United States, with about 10% and 20%, respectively.

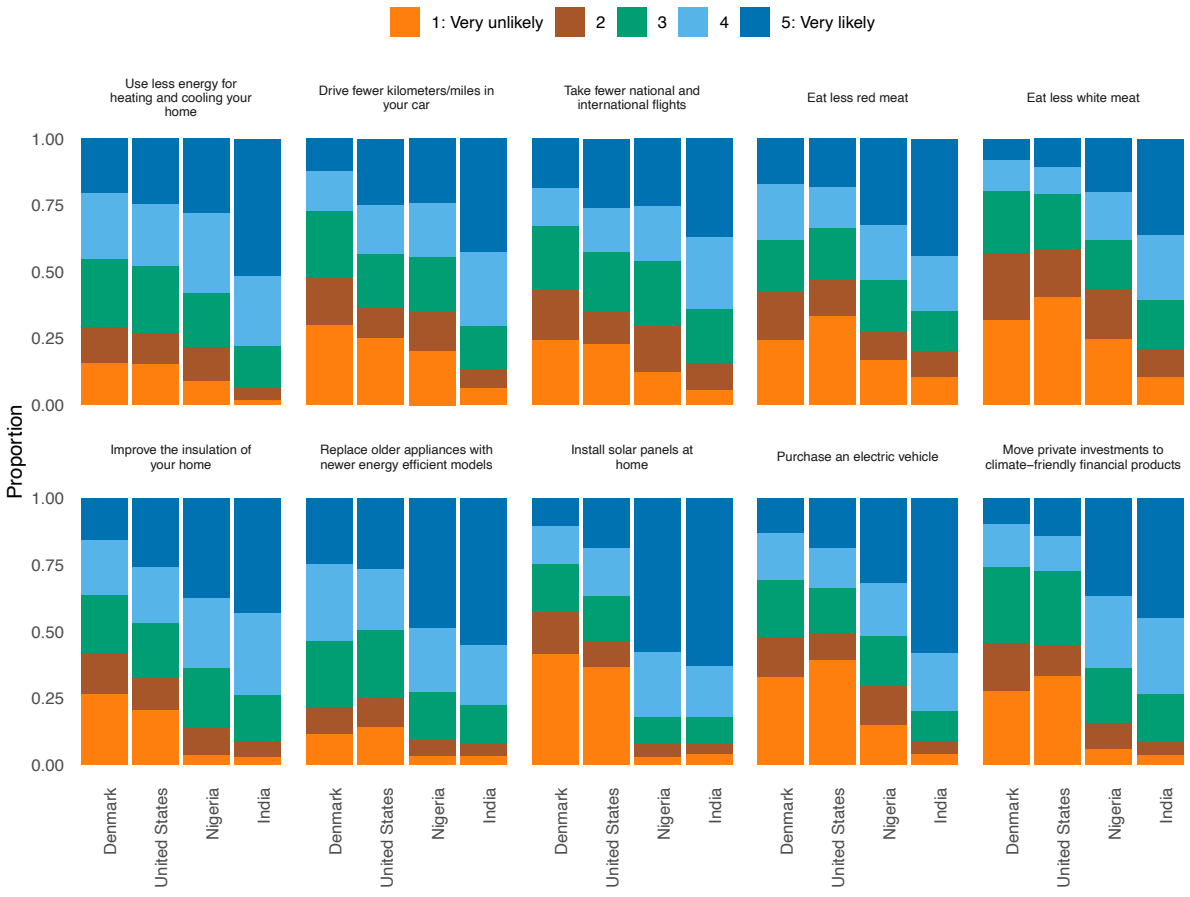


Fig. 1. Perceived behavioral plasticity across behaviors and countries. Proportions of

perceived behavioral plasticity across curtailment (top) and investment (bottom) behaviors and countries.

Perceived behavioral plasticity across income groups

We next used Bayesian regression models to investigate whether perceived behavioral plasticity varied across income groups and whether this relationship differed across countries and behaviors (Fig. 2). We measured income in 15 categories, with the top six categories corresponding to the top 10% income group in the respective countries (see Methods). Across all countries, we found weak evidence against a relationship between income and perceived behavioral plasticity for using less energy for heating and cooling and installing solar panels (Bayes factors (BFs_{01}) 4.2 and 6.5; see Methods). However, we found strong evidence that higher incomes were associated with higher perceived behavioral plasticity across all countries for eating less red meat, purchasing an electric vehicle, and moving private investment to climate-friendly financial products (log BFs_{10} 25.3, 40.5, 6). For taking fewer national and international flights and eating less white meat, we found strong evidence that the relationship between income and perceived behavioral plasticity differed across countries (BFs_{10} 5653 and 865). In particular, participants with higher incomes from Denmark and the United States reported being less likely to take fewer flights. In Denmark, they also reported being less likely to eat less white meat. We found weaker evidence for differences across countries for driving fewer kilometers (BF_{10} 3.6), improving home insulation (BF_{10} 4.6), and replacing older appliances with newer energy-efficient models (BF_{10} 17.2). Differences in mean perceived behavioral plasticity between participants in the top 10% income and the general population mirror the results here (Supplementary Fig. 3). The Bayes factors are robust to different prior specifications (Supplementary Fig. 4 and 5).

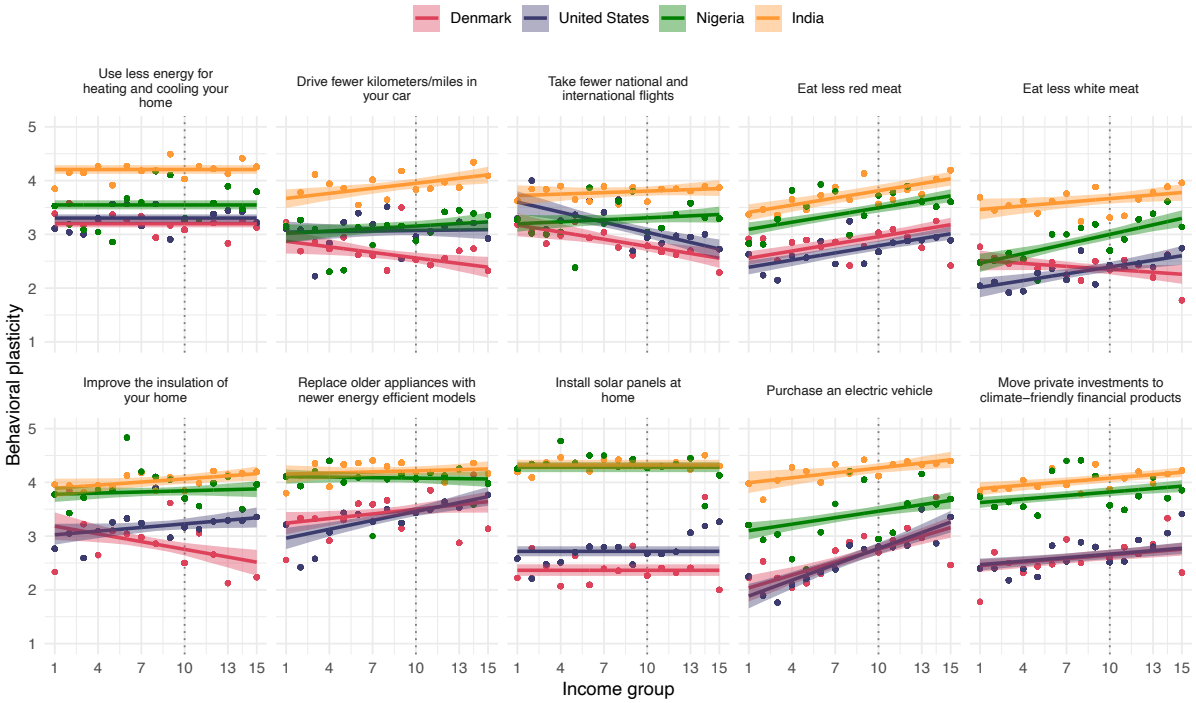


Fig. 2. Relationship of income and behavioral plasticity across behaviors and countries. Posterior mean and 95% credible intervals of the relationship between mean behavioral plasticity and income groups across actions and countries based on the model that best

predicts the data. Points indicate empirical means within income groups. The dotted vertical line indicates the threshold of belonging to the top 10% of income.

There were also notable differences in the proportion of participants who reported already performing the behaviors across income groups (Extended Data Fig. 2; Methods). There was strong evidence across countries (log BFs_{10} from 7.40 to 52.3) that low-income participants were more likely never to fly and not own a car, except for India (BFs_{10} 0.29 and 0.18). Across all countries, we found evidence against an income difference in reporting not heating or cooling the home and not eating red meat (BFs_{01} ranging from 2.76 to 1036.70). There were no income differences regarding not eating white meat except for Nigeria (BF_{01} 31.13), where lower-income participants reported not eating white meat more frequently.

The pattern was often reversed for the investment behaviors. There was strong evidence that high-income groups more often reported already having improved home insulation in Denmark (log BF_{10} 22.60) and the United States (log BF_{10} 6.47), whereas there was strong evidence for no income differences in India (BF_{01} 26.50) and Nigeria (BF_{01} 112.97). For replacing older appliances with newer ones, we found strong evidence for income differences in Denmark (BF_{10} 26.45). We found interesting income differences in Nigeria, where fewer participants in the middle-income category reported having replaced older appliances and installed solar panels. The evidence for income differences regarding installing solar panels was weak for Denmark and the United States, with strong evidence against income differences in India (BF_{01} 87.98). There was strong evidence that participants with higher incomes reported already having purchased an electric vehicle in Denmark (log BF_{10} 25.19) and that there were no income differences in Nigeria (BF_{01} 92.43), with equivocal evidence for the United States and India. Lastly, there was evidence against income differences related to already having moved private investments to climate-friendly financial products across all countries (BFs_{01} ranging from 4.56 to 389.97).

Perceived behavioral plasticity and climate policy support

We next investigated whether the perceived plasticity of a behavior is associated with support for a domain-matched policy and whether that relationship differs across income and countries (Fig. 3). We found strong evidence that higher perceived behavioral plasticity is associated with higher policy support across the three domain-matched policies: taking fewer national and international flights and increase taxes on air travel (log BF_{10} 203); eating less red meat and increase taxes on red meat (log BF_{10} 345); and moving private investment to climate-friendly financial products and mandating banks and investment companies to disclose their greenhouse gas emissions to consumers (log BF_{10} 233). We also found strong evidence that this relationship varies across countries (BFs_{10} 631, 129, 7160; Fig. 3).

Further, we found moderate to strong evidence that the higher the income, the stronger the relationship between perceived behavioral plasticity and matched policy support across all behaviors and countries (BFs_{10} 11.3, 4.4, 6456), and strong evidence that this interaction does not differ across countries (BFs_{01} 233, 67.8, 208.7). We conducted several sensitivity analyses and found that all Bayes factors are robust to different prior specifications (Supplementary Fig. 5).

We observed a similar pattern when examining the relationship between perceived behavioral plasticity and climate policy support for more loosely domain-matched policies, such as driving fewer kilometers and supporting the expansion of public transport (Supplementary Fig. 7-8). Additionally, we found a consistent pattern when averaging perceived behavioral plasticity across all behaviors and policy support across all policies (Supplementary Fig. 9). We also conducted pairwise correlations between all behaviors and

climate policies across countries (Supplementary Fig. 10-13). The results indicate that climate policy support correlated more strongly with perceived plasticity for domain-relevant behaviors compared to those that are not, underscoring the robustness of the finding that perceived behavioral plasticity is strongly associated with domain-matched climate policy support.

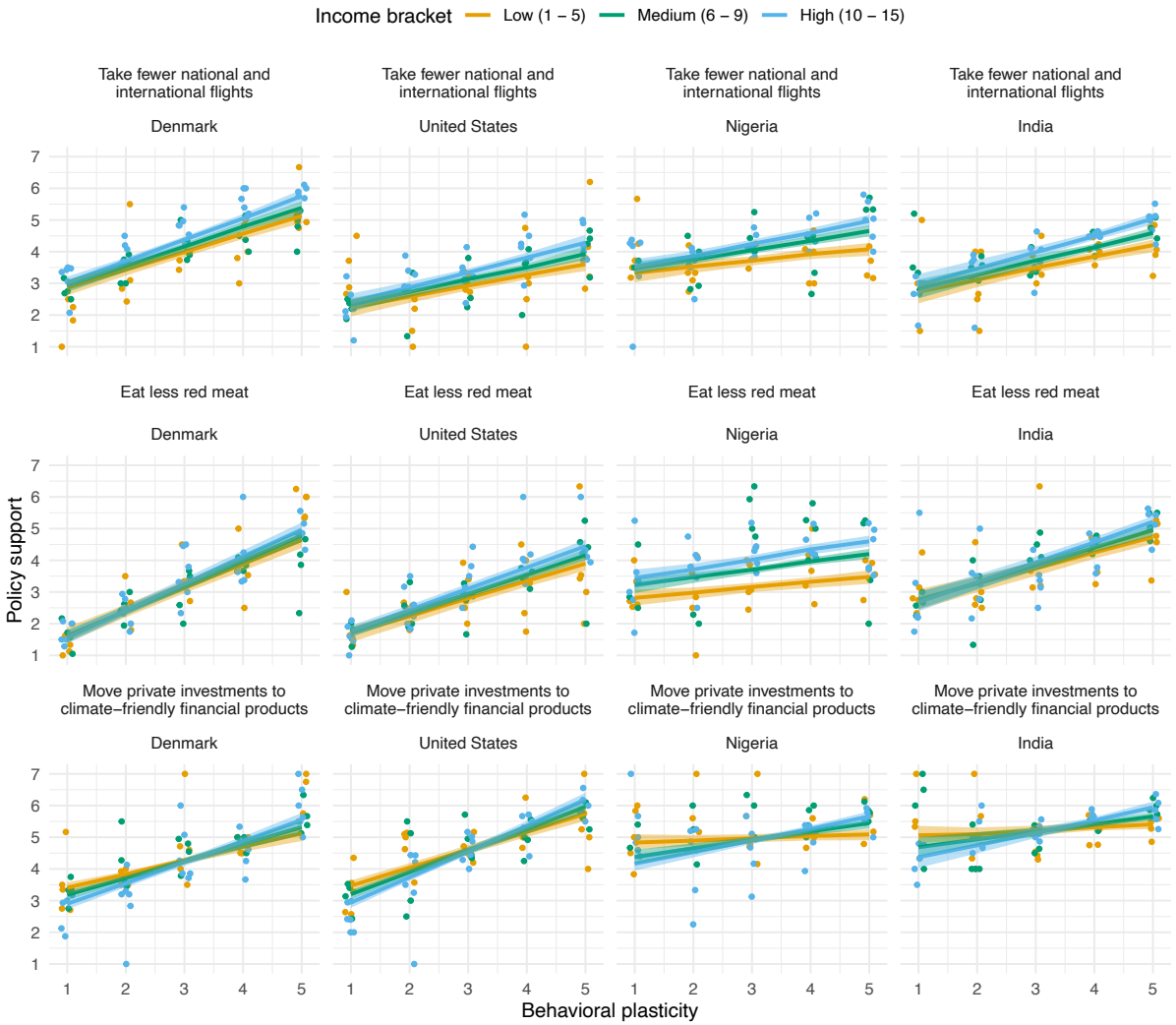


Fig. 3. Relationship between perceived behavioral plasticity and domain-matched policy support across income and countries. Posterior mean and 95% credible intervals of domain-matched mean policy support across perceived plasticity of behaviors, income groups, and countries. Points indicate empirical means within income groups and levels of behavioral plasticity. Taking fewer flights was domain-matched with increasing taxes on air travel, eating less red meat with increasing taxes on red meat, and moving private investment to more climate-friendly financial products by requiring banks and investment companies to disclose their greenhouse gas emissions to consumers.

Discussion

This study examined income differences in perceived behavioral plasticity and its link to climate policy support across four diverse countries. Our results reveal substantial heterogeneity in the perceived plasticity of investment and curtailment behaviors across the countries, with higher plasticity generally observed in India and Nigeria. Additionally, we

identified varying relationships between income and perceptions of behavioral plasticity across different behaviors and countries. Consistent with previous research[12], we also found positive relationships between perceived behavioral plasticity and support for domain-matched climate policies.

Our study's strong representation of high-income individuals permitted a unique examination of potential differences in perceived behavioral plasticity along the income distribution. Interestingly, we found no consistent overall relationship between income and perceived behavioral plasticity but rather substantial heterogeneity across behaviors and countries. Income was positively associated with perceived behavioral plasticity for some behaviors, such as eating less red meat and purchasing an electric vehicle, across all countries but with no evidence for a relationship with the perceived plasticity of using less energy for heating and cooling and installing solar panels at home. For other behaviors, such as taking fewer flights and replacing older appliances with newer energy-efficient models, the relationship between income and perceived behavioral plasticity differed across countries. For example, while participants with higher incomes in Denmark and the United States were less likely to take fewer flights in the future, there was no strong relationship in Nigeria and India. This finding is particularly interesting since air travel is considerably more common among the wealthy and usually makes up a dominant share of their carbon footprints[36,37,42,43].

The heterogeneous results for income highlight its complex role in shaping perceptions of behavioral plasticity. Unlike previous behavior-focused research[32], high-income participants did not consistently report higher perceived plasticity for investment behaviors compared to curtailment behaviors. This inconsistency likely stems from various factors that may differ across countries. One potential overarching reason could be the presence of structural barriers and skewed incentives, which may make certain climate-friendly behavioral changes challenging, even for the wealthy in Denmark and the United States. The results from India and Nigeria are particularly complex to interpret, potentially reflecting a wider knowledge gap in climate action, supportive policy design, response bias, and higher aspirations for investment behaviors.[44]. Further research is required to replicate and unpack these findings.

Evidence of perceived behavioral plasticity can help tailor and target behavior change initiatives to specific population segments (see Ref [45–47] for overview and evidence of behavior change interventions). For example, realizing behavior change among people with low perceived plasticity may require more resources and well-designed structural policies. In contrast, fewer resources and initiatives may be required to realize behavior changes among people reporting high perceived plasticity. Our main analyses focused on participants who could perform the behaviors to represent the pool of realizable behavior changes best. However, this pool is dynamic and can change substantially over time. Ideally, an increasing number of people would adopt low-carbon behaviors, but people can, in principle, also adopt or revert to carbon-intensive behaviors, such as eating red meat, traveling by air, or purchasing a fossil-fuel vehicle. Systematically collecting and curating data on climate-relevant behavior will help effectively monitor changes in behavior and perceived behavioral plasticity across population segments and over time[3].

Replicating prior work[12], we found strong evidence that higher perceived behavioral plasticity is associated with higher climate policy support. Interestingly, we find moderate to strong evidence that the plasticity-support relationship is stronger among high-income compared to lower-income participants. This finding contrasts with earlier qualitative research[31]. We can only speculate about possible explanations for this moderator effect, as our data does not allow for exploring the underlying mechanisms nor identifying the causal structure of our three variables of interest. While confounding remains a potential issue, we see it as implausible that fundamental income differences between people are caused by their

differences in domain-specific policy support or perceptions of behavioral plasticity. Therefore, one potential account could be that income positively affects the perceived cost-to-benefit ratio of a given policy[48]. Specifically, the resourcefulness of high-income individuals may allow them to focus more strongly on the potential benefits of a given climate policy without being as concerned about personal costs. As a result, and all else being equal, they might experience less ambivalence towards more ambitious climate policies and see their consequences as more manageable. However, more research is needed to assess the cognitive and emotional reactions to ambitious climate policies along the socioeconomic spectrum.

Our study has several limitations. Firstly, our sampling strategy deliberately prioritized a strong representation of high-income individuals, who typically have larger personal carbon footprints[24,25] and greater political, organizational, and social influence[15,18,49], over representativeness. This decision resulted in samples skewed from the general population, with a larger share of participants potentially being atypical or outliers in each country. Secondly, perceived plasticity was unexpectedly high for many behaviors in India and some in Nigeria, and these descriptive results should therefore be interpreted with caution. While these perceptions may reflect actual plasticity, we speculate that other factors, such as aspiration, future-oriented actions, psychological distance, or social desirability bias, may have played a role. This is particularly relevant for certain investment behaviors like shifting private investment to climate-friendly financial products, purchasing an electric vehicle, or installing solar panels, where availability and affordability may pose challenges. Although sampling bias in survey research across the Global South remains an ongoing challenge, the diverse representation of high-income individuals from India and Nigeria is an important feature of our study. However, this representation requires careful contextualization for generalizability[50,51], and our results provide only an initial understanding of these complex behavioral dynamics and motivations. We encourage future research to complement our work with mixed-methods research and semi-structured interviews. Thirdly, despite following conventional measurement practices, the climate policy support measures did not fully capture each country's unique climate policy landscape. Finally, although our perceived behavioral plasticity measure is efficient and has notable strengths, it cannot distinguish between motivation and efficacy beliefs. Consequently, we cannot determine whether low perceived behavioral plasticity is due to low motivation to engage in climate-friendly behaviors or external factors that hinder their performance.

In this study, we primarily focused on income-related differences in the perceived plasticity of consumer behaviors and their links to climate policy support. However, people can also effectively promote or counteract climate change mitigation in other areas of their lives[3,52]. The opportunity space and impact potential are particularly large for those with high socioeconomic status[18]. Understanding whether wealthier individuals are aware of and willing to leverage this potential is an important avenue for future research[16].

Methods

We conducted an online survey in Denmark, India, Nigeria, and the United States. Participants were recruited via the market research companies Nielsen (Denmark and Nigeria) and Qualtrics (India and the United States) and received financial compensation. The survey was administered in English, except in Denmark, where it was administered in Danish. In India, only participants who reported feeling comfortable answering in English completed the survey. A total of 4,003 participants completed the survey with the following country breakdown: Denmark (n = 1,001), India (n = 1,001), Nigeria (n = 1,001), and the United States (n = 1,000). In each country, the sampled participants were equally split across two income groups: participants whose personal income fell into the top 10% income bracket

and participants whose personal income was below the threshold for the top 10% income bracket. We used the following income thresholds for the Top 10%[53–55]: Denmark (650.000 DKK), India (300,000 INR), Nigeria (35,000,000 NGN), and the United States (130,000 USD). We adopted this recruitment strategy to ensure a higher representation of high-income individuals, thus deliberating trading off the representativeness of the survey. The survey design, materials, and data exclusion criteria were pre-registered via the Open Science Framework (OSF; https://osf.io/j8txq/). The dataset has been used for another study[56], but the current research questions, reported analyses, and conclusions do not overlap with the other research.

We automatically screened out participants if one or more of the following criteria were met: (1) reported being under 18 years old; (2) felt uncomfortable answering the survey in English (only in India); (3) reported 'prefer not to answer' on the income question; (4) failed the attention check; (5) answered the comprehension check for the concept of personal carbon footprint wrongly twice (a concept only used in another study[56]). Although not pre-registered, we also excluded participants who were likely bots based on their Recaptcha score (n = 10) or who completed the survey unreasonably quickly (n = 180) or more than once (n = 1). These participants were subsequently replaced with other responses by Nielsen and Qualtrics.

In the full sample, the mean age was 42.35 (SD = 16.59), and 56.8% identified as male, 42.8% as female, 0.3% as non-binary, and .01% preferred not to say. An overview of descriptive statistics for each country is presented in Supplementary Table 1.

Measures

Perceived behavioral plasticity. Mirroring recent studies [11,12], participants reported their perceived behavioral plasticity of five investment behaviors and five curtailment behaviors. For both behavioral clusters, participants read the following instruction: "Taking into consideration the COSTS AND INCONVENIENCE for each action, how likely is it that YOU PERSONALLY will take the following actions within the next 1-2 years?" We measured the following investment behaviors rated on a 7-point scale (0 = *I cannot do this*, 1 = *very unlikely*, 5 = *very likely*, 6 = *I have already done this*): (1) purchase an electric vehicle; (2) replace older appliances with newer energy efficient models (e.g., refrigerators, air conditioner); (3) improve the insulation of your home; (4) move private investments to climate-friendly financial products (e.g., free from fossil fuels); and (5) install solar panels at home. The curtailment behaviors, rated on a 6-point scale (1 = *very unlikely*, 5 = *very likely*, 6 = *I don't [behavior]*), were (1) take fewer national or international flights, (2) eat less red meat (e.g., beef, lamb, veal), (3) eat less white meat (e.g., chicken, pork), (4) drive fewer kilometers in your car, and (5) use less energy for heating or cooling your home. For both investment and curtailment behaviors, the perceived behavioral plasticity measure was created from response options 1-5 (1 = *very unlikely*, 5 = *very likely*), thereby excluding participants that either could not perform the behavior (e.g., due to structural constraints) or already performed the behavior.

Climate policy support. Participants indicated their support for 12 prospective climate policies inspired by recent cross-country research [33]. Due to the countries' heterogeneous political systems and policy landscapes, we carefully phrased the policies to ensure applicability across the four countries. The instructions for the policy support items read: "Many countries have introduced new policies to reduce the risk of climate change. This includes policies that require or create incentives for reductions in greenhouse gas emissions across domains and actors. How much do you support or oppose adopting the following policies in [country]?" Responses were assessed on a 7-point Likert scale (1 = *strongly*

oppose, 7 = *strongly support*). The policies were: (1) increase or introduce taxes on products and services that are made from or use fossil fuels (e.g., coal, oil, gas); (2) expand public transport (buses, trams, trains); (3) increase the price of electricity consumption during peak times; (4) increase subsidies for renewable energy projects (e.g., wind and solar energy); (5) strengthen requirements for energy efficiency in buildings; (6) mandate banks and investment companies to reveal their greenhouse gas emissions to consumers; (7) increase or introduce taxes on red meat (e.g., beef, lamb, veal); (8) increase or introduce taxes on air travel; (9) introduce a mandatory carbon footprint label on consumer products; (10) ban the sale of diesel and petrol-engine cars; (11) increase subsidies for technologies that remove greenhouse gases from the atmosphere; and (12) increase subsidies for food products with low greenhouse gas emissions (e.g., fruit, vegetables, legumes, cereals).

Income. We measured participants' personal income by asking, "What was your total annual personal income before taxes in 2022?" rated on a 10-point scale (plus a 'prefer not to answer' option). The answer options were adapted to each country, with response option 10 being the income threshold for belonging to the Top 10% of income (e.g., "*₹300,000 or more*"). Participants who reported an income belonging to the Top 10% were subsequently asked another 6-point income question to assess their personal income more accurately: "You indicated that your total annual personal income before taxes in 2022 was more than [top 10% income threshold]. Please select the band that most precisely describes your income." We subsequently created a combined income measure with 15 response categories. The survey also included other measures not detailed here (see the pre-registration for further details).

Analytical strategy

We used Bayesian inference to assess the evidence for the presence or absence of effects[57,58]. Specifically, for results shown in Fig. 2, we used the R package *BayesFactor*[59] to estimate linear models for each climate-relevant behavior, with behavioral plasticity as the outcome variable and income, country, and their interaction as predictor variables. We assigned a default (zero-centered) Jeffreys-Zellner-Siow priors with scale $\sqrt{\frac{1}{2}}$ (0.7071) to the effect of country and $\sqrt{\frac{1}{2}}$ / 4 (0.1768) to the effects of income and the interaction, given that we a priori expected smaller effects for continuous variables than for country[60]. These priors refer to standardized effect sizes for the continuous variables. For example, a prior scale of 0.1768 for income means that we expected that 50% of the probability mass for the effect of a one standard deviation increase in income is within a 0.1768 standard deviation increase or decrease in behavioral plasticity (with the mean being a 0 increase). For the effect of country, a scale of 0.7071 means that the difference between two countries in mean perceived behavioral plasticity is, with 50% probability, between -0.7071 and 0.7071. We compared five models that included: (1) only an intercept, (2) only the main effect of income, (3) only the main effect of country, (4) both main effects, and (5) both main effects and their interaction. We used Bayes factors to assess the strength and evidence for or against an effect, yet it is not obvious which models to compare. For example, to assess the evidence for the effect of income, one might be tempted to compare the model that includes only the main effect of income against the intercept-model only, or compare the model that includes both income and country against a model that includes only country. The problem of selecting a reference model becomes more acute with more variables. Instead, we used model-averaging and inclusion Bayes factors, which avoid this problem and take the uncertainty across models into account[61,62]. Inclusion Bayes factors are defined as the ratio of posterior odds that a particular effect is present in the data compared to its prior odds. We assumed a uniform prior

over models, but our results are robust to different choices (e.g., the popular beta-binomial prior with a = b = 1).

Take the example of income again. First, we calculated the prior odds that income is included as an effect; that is, we divided the prior probability that income is included in a model by the prior probability that it is not included in a model while removing from the set of models those that include an interaction including income, as this would change its meaning. In this case, where we only have five models, the prior odds for income are 1. The procedure is the same for calculating the posterior odds, except we used posterior model probabilities. The inclusion Bayes factor is then the ratio of the posterior odds to the prior odds. When assessing the evidence for an interaction, we only use those models that include the interaction and those that do not include it but include all relevant main effects[62]. Fig. 2 shows the regression lines associated with the model with the highest marginal likelihood, that is, the model that best predicted the data.

We used a similar approach for the results shown in Fig. 3, estimating linear models with the domain-matched climate policy as the outcome variable and the domain-matched perceived behavioral plasticity, income, and country as predictors. We again assigned JZS priors with scale $\sqrt{\frac{1}{2}}$ to the effect of country and $\sqrt{\frac{1}{2}} / 4$ to the effect of perceived behavioral plasticity and income, as well as any interaction. We estimated models that include only an intercept, only the main effects (individually or jointly), main effects and two-way interactions, and main effects and two-way interactions and the three-way interaction. For each model that had an interaction, we specified that the respective main effect must feature too. This yielded a total of 19 different models. We again used model-averaging to calculate inclusion Bayes factors, taking the uncertainty across all models into account. Again, we assigned a uniform prior to the models, but our results are robust with respect to a different choice. Fig. 3 shows the regression lines associated with the models with the highest marginal likelihood, that is, those that predict the data best.

We conducted the Bayesian proportion tests (Extended Data Fig. 2) using the *BayesFactor* R package using a default Beta(1, 1) prior and assumed an independent multinomial sampling plan[63].

Data availability
The data and study materials are available via the Open Science Framework: https://osf.io/j8txq/

Code availability
The analysis code is available via the Open Science Framework: https://osf.io/j8txq/

Acknowledgments
This work was supported by the Carlsberg Foundation, grant number CF22-1059 (to K.S.N), the Bill and Melinda Gates Foundation, grant OPP1144 (to C.A.E and R.D.), Cambridge Humanities Research Grant (to R.D.), Keynes Fund for Applied Economics [JHVH] (to R.D.), and the Swiss National Science Foundation - Eccellenza Professorial Fellowship, grant number PCEFPI_203283 (to U.J.J.H).

Ethics declaration
The survey was approved by the IRB at the University of Basel (020-22-1). We obtained informed consent from all participants before completing the survey.

Author contributions
K.S.N., F.D., J.M.B., and W.H. designed the research with support from R.D., C.A.E., S.G., W.G., U.J.J.H.. F.D. analyzed the data with support from K.S.N., J.M.B., and W.H.. K.S.N. and F.D. wrote the original draft. R.D., C.A.E., S.G., W.G., U.J.J.H., W.H., and J.M.B. reviewed and edited the manuscript.

Competing interests
The authors declare no competing interests.

References

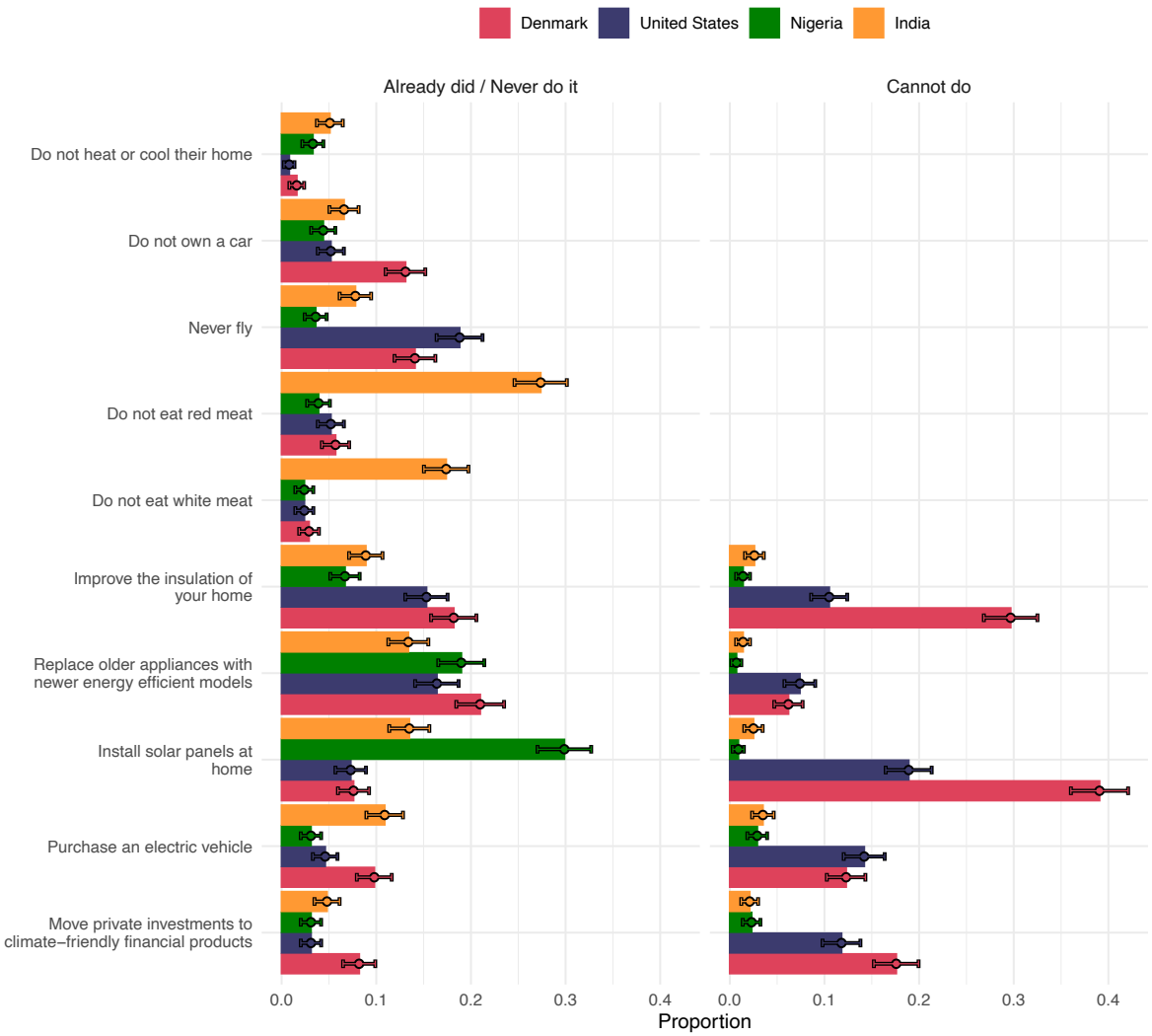
1. Creutzig, F. *et al.* Demand-side solutions to climate change mitigation consistent with high levels of wellbeing. *Nat. Clim. Change* **12**, 36–46 (2022).
2. Creutzig, F. *et al.* Demand, services and social aspects of mitigation. in *IPCC, 2022: Climate Change 2022: Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change* 752–943 (Cambridge University Press, 2022).
3. Nielsen, K. S. *et al.* Realizing the full potential of behavioural science for climate change mitigation. *Nat. Clim. Change* 322–330 (2024).
4. Geels, F. W. Demand-side emission reduction through behavior change or technology adoption? Empirical evidence from UK heating, mobility, and electricity use. *One Earth* **6**, 337–340 (2023).
5. Nielsen, K. S. *et al.* Improving climate change mitigation analysis: A framework for examining feasibility. *One Earth* **3**, 325–336 (2020).
6. Stern, P. C., Dietz, T., Nielsen, K. S., Peng, W. & Vandenbergh, M. P. Feasible climate mitigation. *Nat. Clim. Change* **2023**, 6–8 (2023).
7. Westlake, S., Demski, C. & Pidgeon, N. "We can't be too saintly": Why members of parliament in the United Kingdom are reluctant to lead by example with low-carbon behaviour. *Energy Res. Soc. Sci.* **117**, 103717 (2024).
8. Dietz, T., Gardner, G. T., Gilligan, J., Stern, P. C. & Vandenbergh, M. P. Household actions can provide a behavioral wedge to rapidly reduce U.S. carbon emissions. *Proc. Natl. Acad. Sci.* **106**, 18452–18456 (2009).
9. Stern, P. C. Contributions of psychology to limiting climate change. *Am. Psychol.* **66**, 303–314 (2011).
10. Nielsen, K. S. *et al.* How psychology can help limit climate change. *Am. Psychol.* **76**, 130–144 (2021).
11. Allen, S., Dietz, T. & Mccright, A. M. Measuring household energy efficiency behaviors with attention to behavioral plasticity in the United States. *Energy Res. Soc. Sci.* **10**, 133–140 (2015).
12. Kukowski, C. A. *et al.* The perceived feasibility of behavior change is positively associated with support for domain-matched climate policies. *One Earth* **6**, 1554–1563 (2023).
13. McFadden, B. R., Ferraro, P. J. & Messer, K. D. Private costs of carbon emissions abatement by limiting beef consumption and vehicle use in the United States. *PLoS ONE* **17**, e0261372 (2022).
14. Herberz, M., Hahnel, U. J. J. & Brosch, T. Counteracting electric vehicle range concern with a scalable behavioural intervention. *Nat. Energy* **76**, 503–510 (2022).
15. Dietz, T. & Whitley, C. T. Inequality, decisions, and altruism. *Sociol. Dev.* **4**, 282–303 (2018).
16. Duncan, S., Hjelmskog, A. & Papies, E. K. No agency and responsibility for the privileged? Perceptions of climate action among high socioeconomic status citizens in the United Kingdom. *Energy Res. Soc. Sci.* **115**, 103604 (2024).
17. Kukowski, C. A. & Garnett, E. E. Tackling inequality is essential for behaviour change for net zero. *Nat. Clim. Change* **14**, 2–4 (2023).
18. Nielsen, K. S., Nicholas, K. A., Creutzig, F., Dietz, T. & Stern, P. C. The role of high-socioeconomic-status people in locking in or rapidly reducing energy-driven greenhouse gas emissions. *Nat. Energy* **6**, 1011–1016 (2021).
19. Borenstein, S. & Davis, L. W. The distributional effects of US clean energy tax credits. *Tax Policy Econ.* **30**, 191–234 (2016).
20. Sovacool, B. K., Newell, P., Carley, S. & Fanzo, J. Equity, technological innovation and

sustainable behaviour in a low-carbon future. *Nat. Hum. Behav.* **6**, 326–337 (2022).
21. Babagoli, M. A., Kaufman, T. K., Noyes, P. & Sheffield, P. E. Exploring the health and spatial equity implications of the New York City Bike share system. *J. Transp. Health* **13**, 200–209 (2019).
22. Ermagun, A. & Tian, J. Charging into inequality: A national study of social, economic, and environment correlates of electric vehicle charging stations. *Energy Res. Soc. Sci.* **115**, 103622 (2024).
23. Bruckner, B., Hubacek, K., Shan, Y., Zhong, H. & Feng, K. Impacts of poverty alleviation on national and global carbon emissions. *Nat. Sustain.* **5**, 311–320 (2022).
24. Chancel, L. Global carbon inequality over 1990–2019. *Nat. Sustain.* **5**, 931–938 (2022).
25. Oswald, Y., Owen, A. & Steinberger, J. K. Large inequality in international and intranational energy footprints between income groups and across consumption categories. *Nat. Energy* **5**, 231–239 (2020).
26. Wiedmann, T., Lenzen, M., Keyßer, L. T. & Steinberger, J. K. Scientists' warning on affluence. *Nat. Commun.* **11**, 3107–3107 (2020).
27. Cass, N., Büchs, M. & Lucas, K. How are high-carbon lifestyles justified? Exploring the discursive strategies of excess energy consumers in the United Kingdom. *Energy Res. Soc. Sci.* **97**, 102951 (2023).
28. Moorcroft, H., Hampton, S. & Whitmarsh, L. Climate change and wealth: Understanding and improving the carbon capability of the wealthiest people in the UK. SSRN Scholarly Paper at https://doi.org/10.2139/ssrn.4723179 (2024).
29. De Nardo, M., Brooks, J. S., Klinsky, S. & Wilson, C. Social signals and sustainability: ambiguity about motivations can affect status perceptions of efficiency and curtailment behaviors. *Environ. Syst. Decis.* **37**, 184–197 (2017).
30. Uren, H. V., Roberts, L. D., Dzidic, P. L. & Leviston, Z. High-status pro-environmental behaviors: Costly, effortful, and visible. *Environ. Behav.* **53**, 455–484 (2021).
31. Duncan, S., Hjelmskog, A. & Papies, E. K. A low-carbon future? Individuals with high socioeconomic status in the UK show little willingness to change high-carbon lifestyles for climate change mitigation. Preprint at https://doi.org/10.31234/osf.io/mh8dq (2023).
32. Umit, R., Poortinga, W., Jokinen, P. & Pohjolainen, P. The role of income in energy efficiency and curtailment behaviours: Findings from 22 European countries. *Energy Res. Soc. Sci.* **53**, 206–214 (2019).
33. Dechezleprêtre, A. *et al.* Fighting climate change: International attitudes toward climate policies. Preprint at https://www.nber.org/papers/w30265 (2022).
34. Maki, A. *et al.* Meta-analysis of pro-environmental behaviour spillover. *Nat. Sustain.* **2**, 307–315 (2019).
35. Kukowski, C. A., Bernecker, K., Nielsen, K. S., Hofmann, W. & Brandstätter, V. Regulate me! Self-control dissatisfaction in meat reduction success relates to stronger support for behavior-regulating policy. *J. Environ. Psychol.* **85**, 101922 (2023).
36. Ivanova, D. & Wood, R. The unequal distribution of household carbon footprints in Europe and its link to sustainability. *Glob. Sustain.* **3**, 1–12 (2020).
37. Otto, I. M., Kim, K. M., Dubrovsky, N. & Lucht, W. Shift the focus from the super-poor to the super-rich. *Nat. Clim. Change* **9**, 82–84 (2019).
38. Ravallion, M. Missing top income recipients. *J. Econ. Inequal.* **20**, 205–222 (2022).
39. Brick, C. *et al.* Current research practices on pro-environmental behavior: A survey of environmental psychologists. *J. Environ. Psychol.* **97**, 102375 (2024).
40. Ghai, S. It's time to reimagine sample diversity and retire the WEIRD dichotomy. *Nat. Hum. Behav.* **5**, 971–972 (2021).
41. Henrich, J., Heine, S. J. & Norenzayan, A. The weirdest people in the world? *Behav. Brain Sci.* **33**, 61–83 (2010).

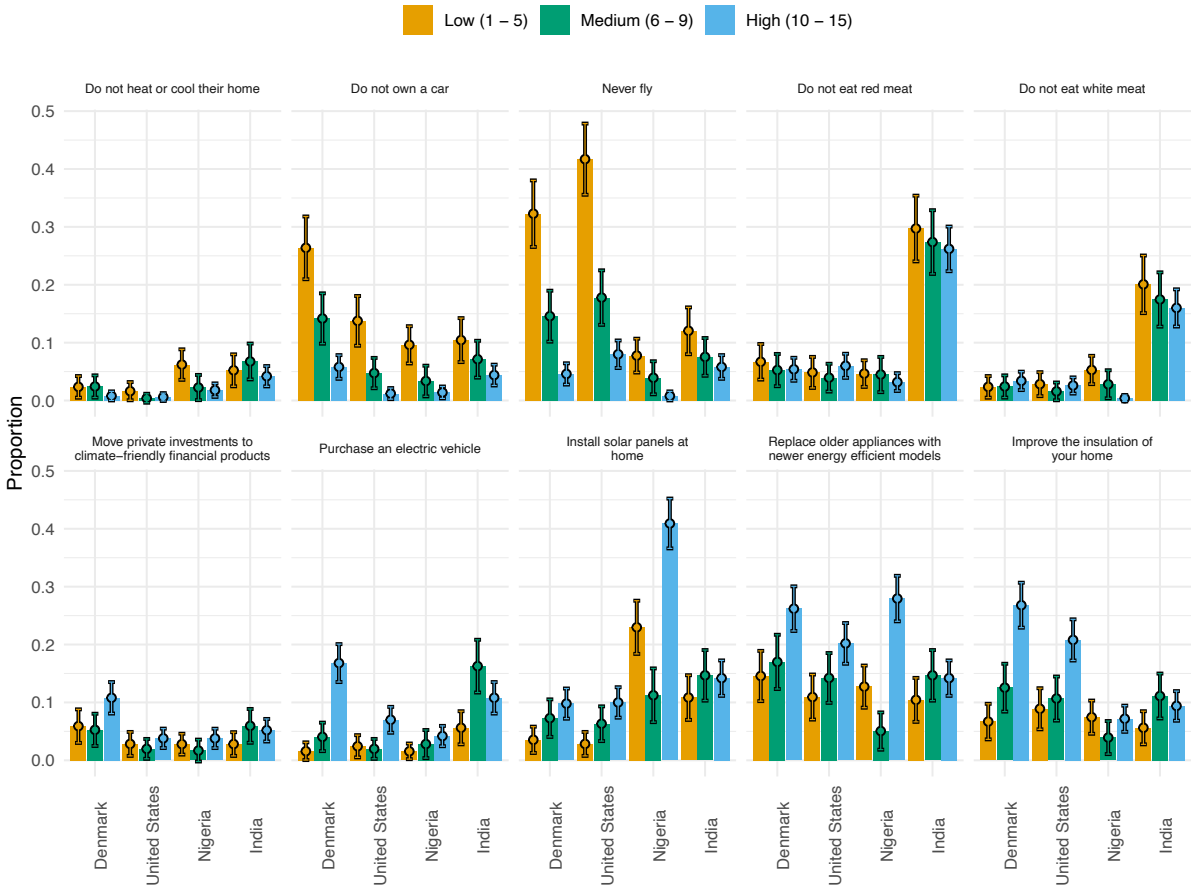
42. Gössling, S. & Humpe, A. The global scale, distribution and growth of aviation: Implications for climate change. *Glob. Environ. Change* **65**, 102194 (2020).
43. Gössling, S. Celebrities, air travel, and social norms. *Ann. Tour. Res.* **79**, 102775 (2019).
44. Marquardt, J., Fünfgeld, A. & Elsässer, J. P. Institutionalizing climate change mitigation in the Global South: Current trends and future research. *Earth Syst. Gov.* **15**, 100163 (2023).
45. Khanna, T. M. *et al.* A multi-country meta-analysis on the role of behavioural change in reducing energy consumption and CO2 emissions in residential buildings. *Nat. Energy* **6**, 925–932 (2021).
46. Composto, J. W. & Weber, E. U. Effectiveness of behavioural interventions to reduce household energy demand: a scoping review. *Environ. Res. Lett.* **17**, 063005 (2022).
47. Bergquist, M., Thiel, M., Goldberg, M. H. & van der Linden, S. Field interventions for climate change mitigation behaviors: A second-order meta-analysis. *Proc. Natl. Acad. Sci.* **120**, e2214851120 (2023).
48. Grelle, S. & Hofmann, W. When and why do people accept public policy interventions? An integrative public policy acceptance framework. *Perspect. Psychol. Sci.* **19**, 258–279 (2024).
49. Green, F. & Healy, N. How inequality fuels climate change: The climate case for a Green New Deal. *One Earth* **5**, 635–649 (2022).
50. Brown, S., Saxena, D. & Wall, P. J. Data collection in the global south: practical, methodological, and philosophical considerations. *Inf. Technol. Dev.* **30**, 379–399 (2024).
51. Boas, T. C., Christenson, D. P. & Glick, D. M. Recruiting large online samples in the United States and India: Facebook, Mechanical Turk, and Qualtrics. *Polit. Sci. Res. Methods* **8**, 232–250 (2020).
52. Hampton, S. & Whitmarsh, L. Choices for climate action: A review of the multiple roles individuals play. *One Earth* **6**, 1157–1172 (2023).
53. CEPOS. Income, distribution, and incentives in Denmark 2019. https://cepos.dk/publications/income-distribution-and-incentives-in-denmark-2019/ (2019).
54. The Times of India. People earning Rs 25,000 per month fall in top 10% of wage earners in India: Report. *The Times of India* https://timesofindia.indiatimes.com/business/india-business/people-earning-rs-25000-per-month-fall-in-top-10-of-wage-earners-in-india-report/articleshow/91694940.cms (2022).
55. PK. Average, median, Top 1% individual income percentiles. *DQYDJ* https://dqydj.com/average-median-top-individual-income-percentiles/ (2023).
56. Nielsen, K. S. *et al.* Underestimation of personal carbon footprint inequality in four diverse countries. *Nat. Clim. Change* 1–8 (2024) doi:10.1038/s41558-024-02130-y.
57. Kass, R. E. & Raftery, A. E. Bayes Factors. *J. Am. Stat. Assoc.* **90**, 773–795 (1995).
58. Wagenmakers, E.-J., Morey, R. D. & Lee, M. D. Bayesian benefits for the pragmatic researcher. *Curr. Dir. Psychol. Sci.* **25**, 169–176 (2016).
59. Schönbrodt, F. D. & Wagenmakers, E.-J. Bayes factor design analysis: Planning for compelling evidence. *Psychon. Bull. Rev.* **25**, 128–142 (2018).
60. Rouder, J. N. & Morey, R. D. Default Bayes factors for model selection in regression. *Multivar. Behav. Res.* **47**, 877–903 (2012).
61. Hinne, M., Gronau, Q. F., Van Den Bergh, D. & Wagenmakers, E.-J. A Conceptual Introduction to Bayesian Model Averaging. *Adv. Methods Pract. Psychol. Sci.* **3**, 200–215 (2020).
62. van den Bergh, D. *et al.* A tutorial on conducting and interpreting a Bayesian ANOVA in JASP. *L'Année Psychol.* **120**, 73–96 (2020).
63. Jamil, T. *et al.* Default "Gunel and Dickey" Bayes factors for contingency tables. *Behav.*

Res. Methods **49**, 638–652 (2017).

Extended Data



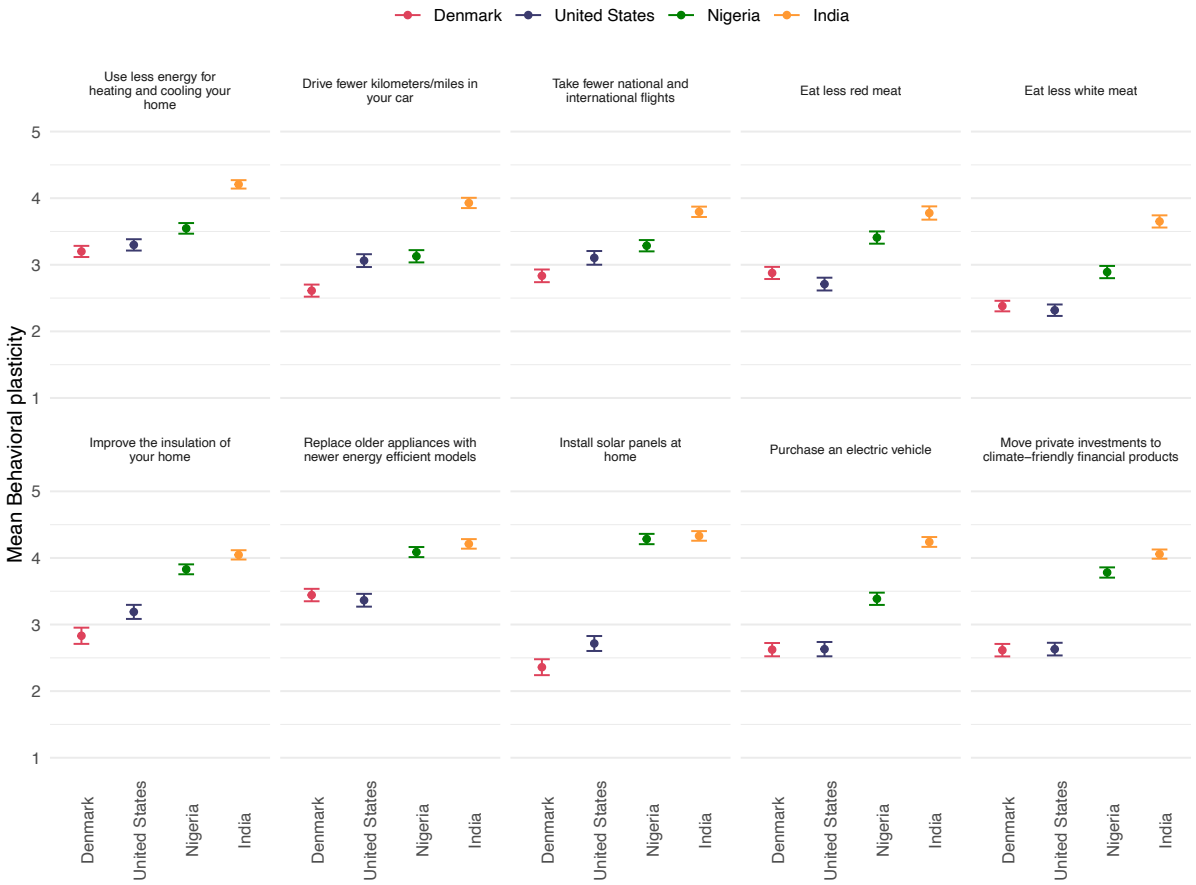
Extended Data Fig. 1. The proportions of participants across countries indicating that they already did or cannot do a specific behavior. The "I cannot do it" response option was only offered for investment behaviors (Methods).



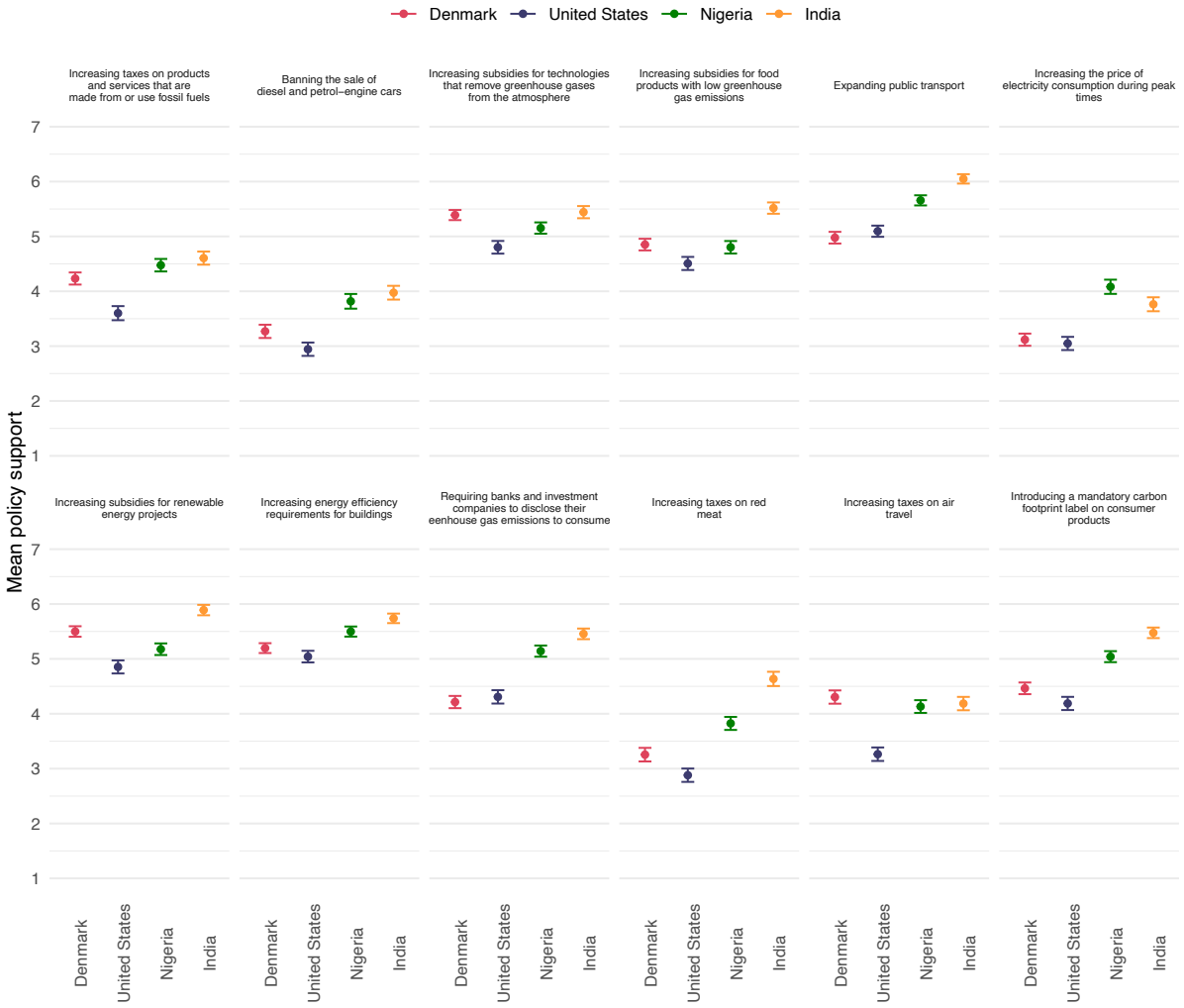
Extended Data Fig. 2. The proportions of participants indicating they already did or never do a particular behavior across income groups and countries. Bayes factor tests using all 15 income groups indicate strong evidence for a difference in proportions for Denmark for the behaviors of never flying, not owning a car, purchasing an electric vehicle, and improving the insulation of one's home; for the United States, only for never flying; for Nigeria for replacing older appliances with newer energy efficient models and installing solar panels at home; and for India there was no evidence of a difference across income groups.

Supplementary information

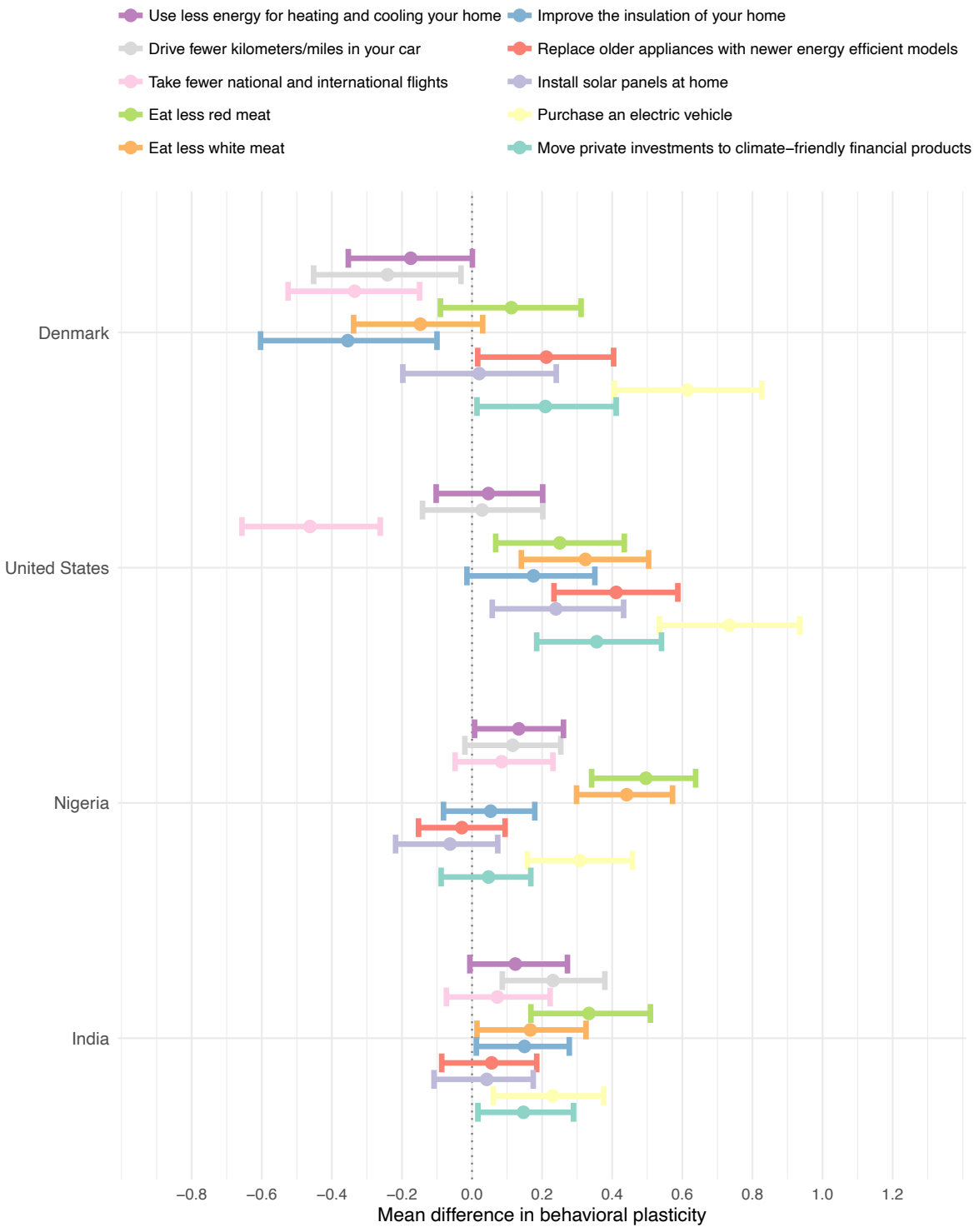
Supplementary Fig. 1. Mean perceived behavioral plasticity across behaviors and countries. Error bars indicate 95% confidence intervals.



Supplementary Fig. 2. Mean climate policy support across behaviors and countries. Error bars indicate 95% confidence intervals.



Supplementary Fig. 3. Differences in mean perceived behavioral plasticity between participants in the top 10% income group and the general population. Points indicate posterior mean difference across items with 95% credible intervals.

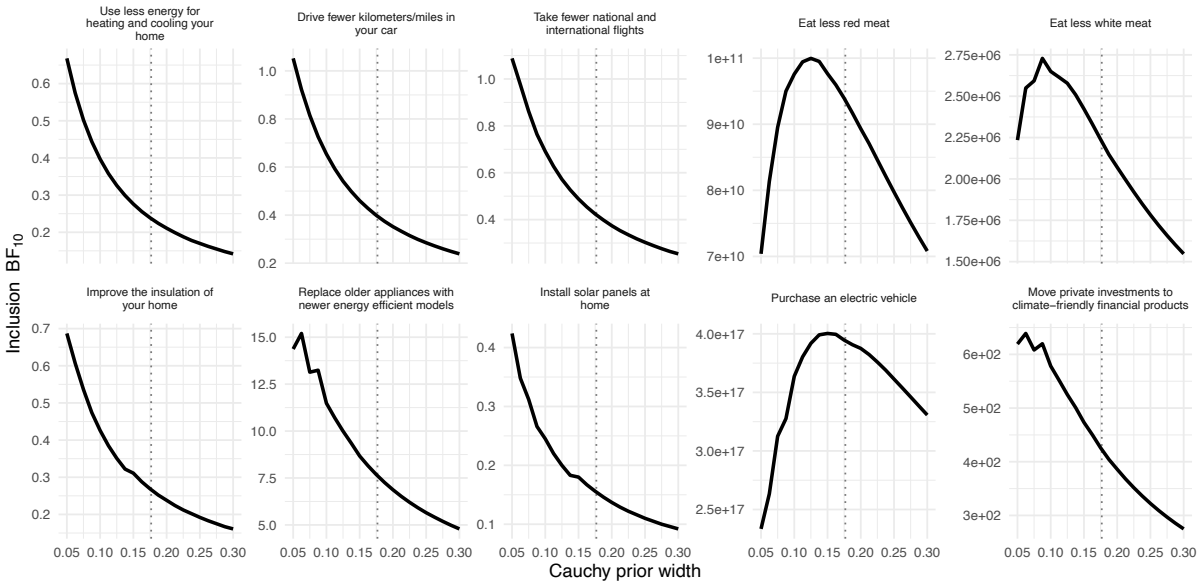


For the main effect of income, we varied the width of the Cauchy prior to assess the sensitivity of the inclusion Bayes factor. At the same time, we vary the width of the Cauchy prior for the interaction effect of income and country accordingly. We found that for eating less red meat, eating less white meat, purchasing an electric vehicle, and moving private investments to climate-friendly financial products, the inclusion Bayes factor in favor of a

main effect of income remains extraordinarily large (Supplementary Fig. 3). They are thus very robust to differences in the prior. For the first three of these behaviors, the inclusion Bayes factor first increases and then decreases, while the Bayes factor only decreases for the last item. This suggests something about their effect size: the effect size for moving private investment is small because models that assume a small effect size predict the data better than models that assume a larger effect size (as encoded in the Cauchy prior width). For the other behaviors, the effect size is larger because models that expect a larger effect size perform better than models that assume a smaller effect size—but only up to a point where models that expect a much larger effect size perform worse again. The evidence for a main effect of income on the behavior of replacing older appliances with newer energy-efficient models remains weak to moderate across prior widths.

The inclusion Bayes factors for all other items were (almost always) smaller than 1, indicating evidence against the respective main effects. As larger effect sizes were expected, that is, as the Cauchy prior width increases, the Bayes factor gets smaller and smaller, suggesting the evidence against the main effects becomes stronger and stronger. However, the inclusion Bayes factor against the main effects (i.e., the reciprocal of the inclusion Bayes factor in favor of them) never reached 10, even for a prior width of 0.30 (except for installing solar panels, where it was slightly above). This indicates the robustness of our main conclusion that there is only weak evidence against a main effect of income for these behaviors.

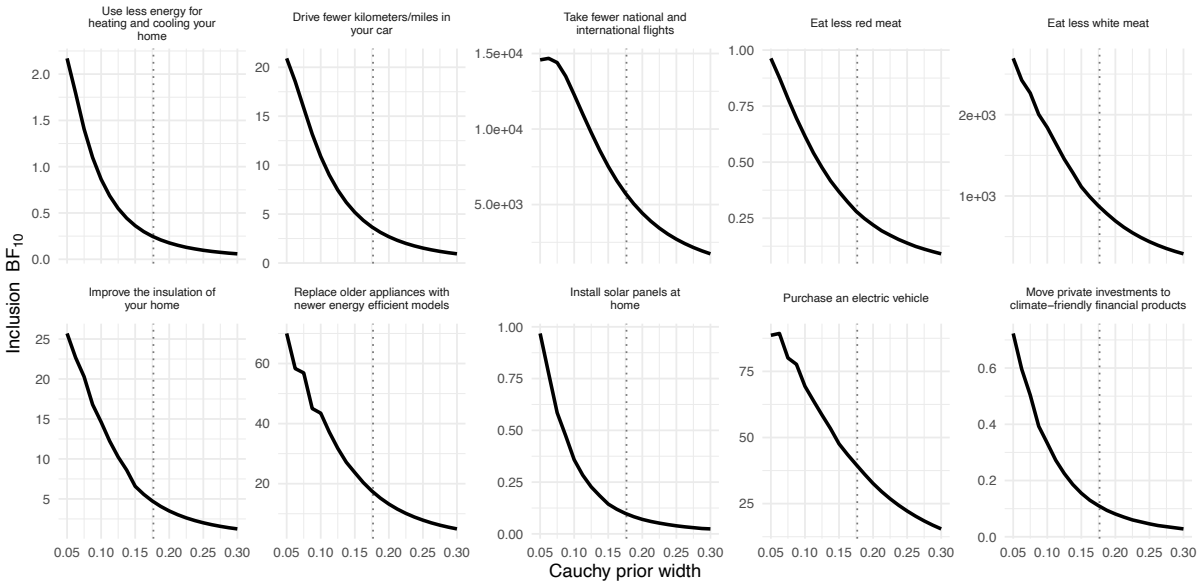
Supplementary Fig. 4. Inclusion Bayes factors for the main effect of income across Cauchy prior scales.



We do the same sensitivity analysis for the interaction effect of income and country (Supplementary Fig. 4). We find that the evidence for such an interaction remains high and robust for taking fewer flights, eating less white meat, purchasing an electric vehicle, and replacing older appliances with newer energy-efficient models. We do see that the Bayes factors decrease markedly as we increase the Cauchy prior width, suggesting that the effect sizes are smaller than for the main effect of income. This is what one might expect, but the *BayesFactor* R package currently does not allow specifying different priors for main effects and interactions. The evidence for the interaction for the behaviors of driving fewer kilometers and improving home insulation moves from moderate Bayes factors of around 20 to around 1, indicating that the evidence for these effects is not very robust to changes in the prior width, although prior widths of 0.30 may also be too unreasonable.

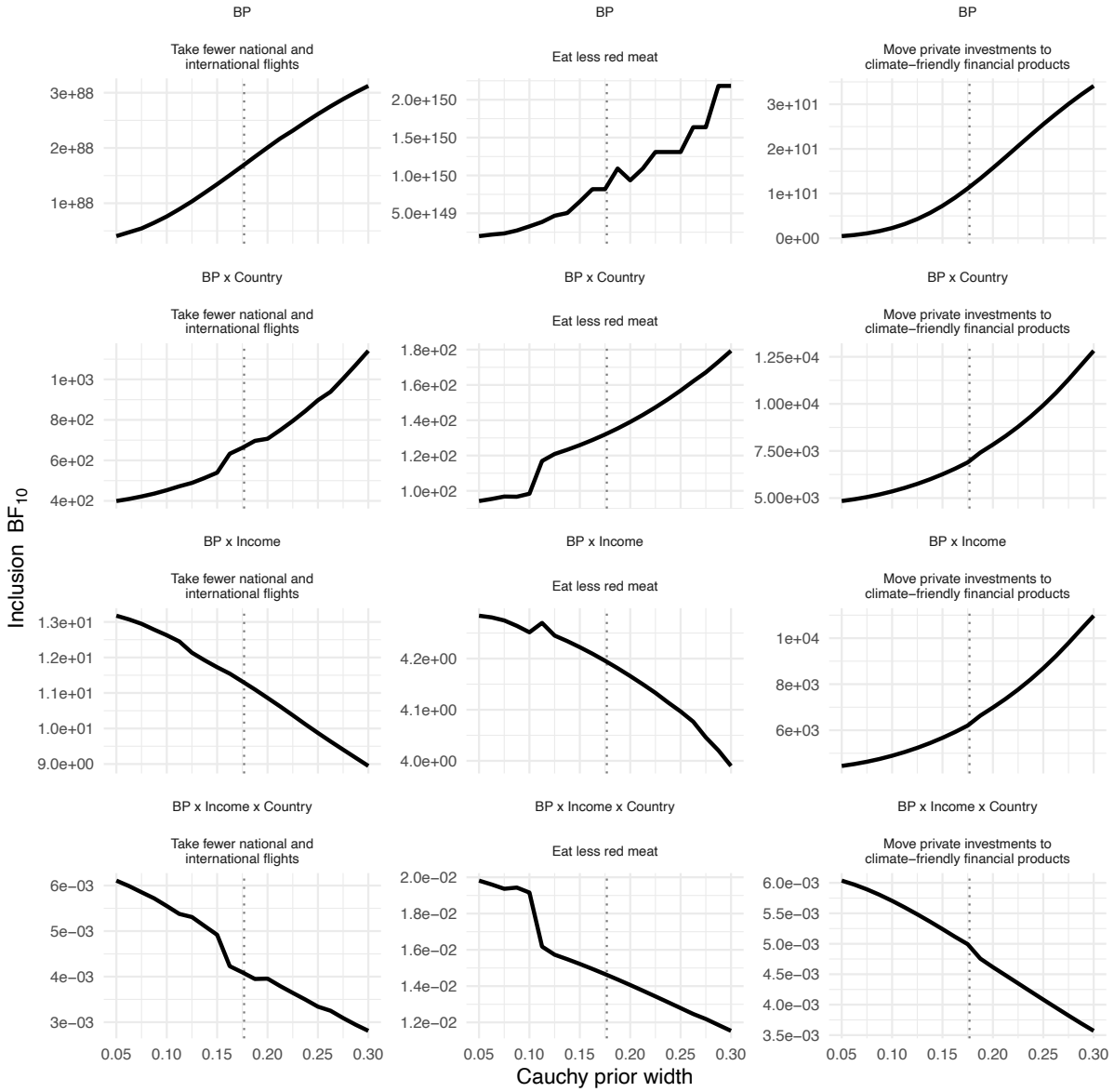
The Bayes factors for all other behaviors were mostly below 1 for much of the prior width parameter space but never exceedingly so (except for very large prior widths). This suggests that the statement of having only weak evidence against such interactions—as reported in the main text—is robust.

Supplementary Fig. 5. Inclusion Bayes factors for the interaction effect of income and country across Cauchy prior scales.

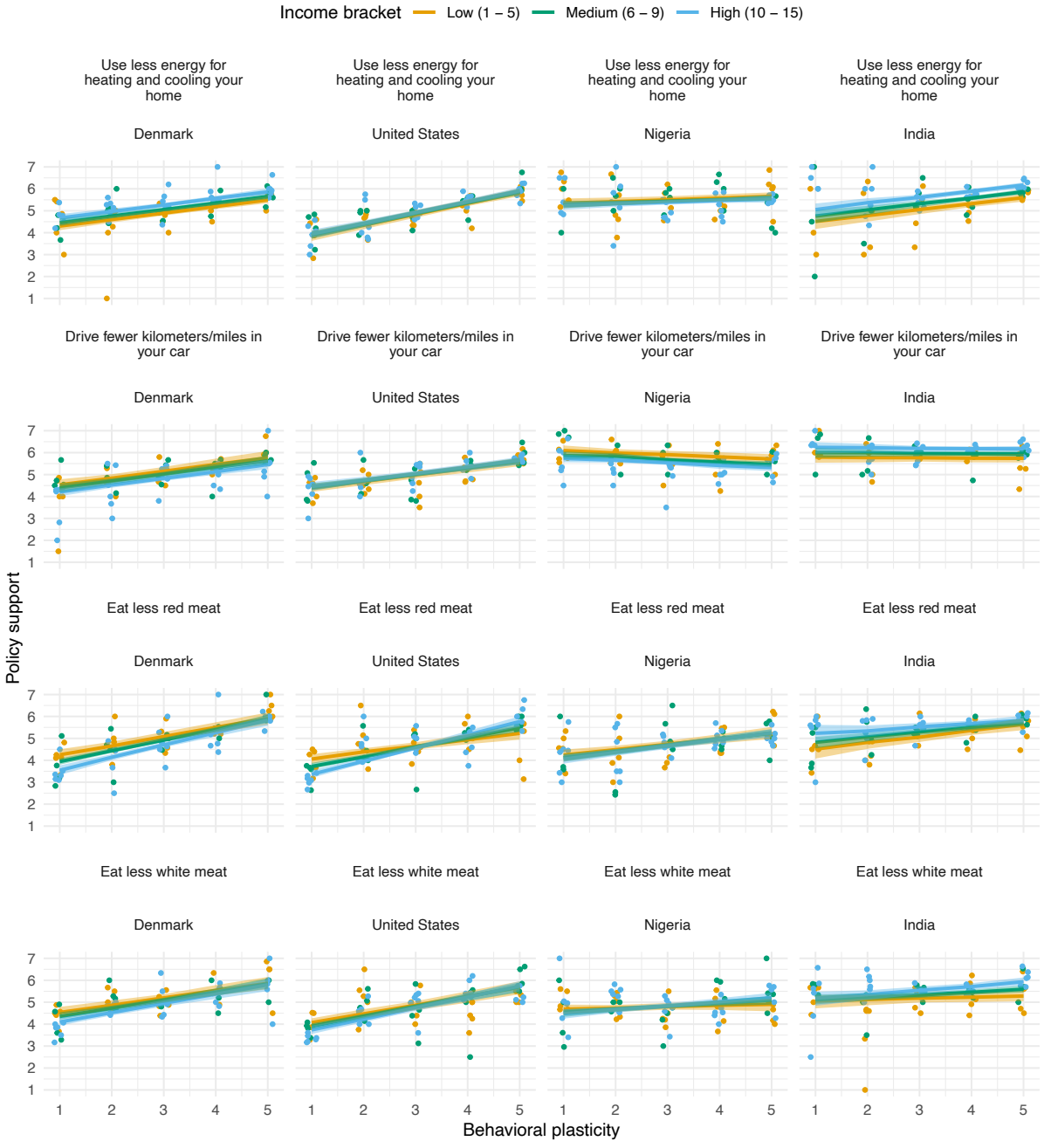


To assess the robustness of the evidence against and for (a) the main effect of perceived behavioral plasticity, (b) the two-way interaction effect of behavioral plasticity and country, (c) the two-way interaction effect of behavioral plasticity and income, and (d) the three-way interaction effect of behavioral plasticity, income, and country, we vary the Cauchy prior width (Supplementary Fig. 5). We find that the evidence for a main effect of perceived behavioral plasticity and for an interaction of perceived behavioral plasticity and country remains overwhelming for all three behaviors. For the interaction of perceived behavioral plasticity and income, we find consistent moderate evidence for taking fewer flights, consistent weak evidence for eating less red meat, and consistent overwhelming evidence for moving private investments. The evidence against a three-way interaction is overwhelming across prior widths for all items except for eating less red meat, for which the evidence is slightly lower but still substantial.

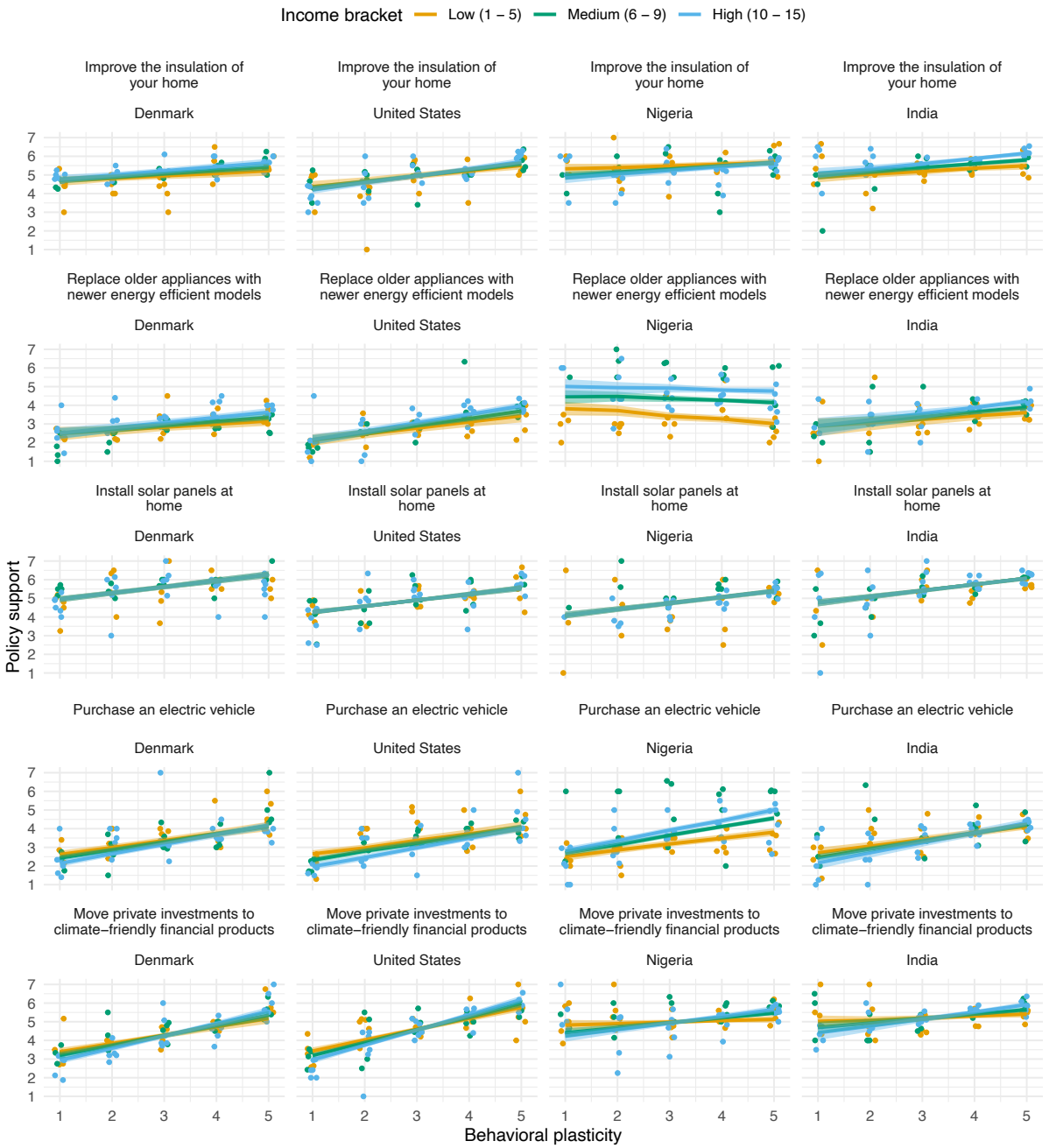
Supplementary Fig. 6. Inclusion Bayes factors for the main effect of perceived behavioral plasticity (first row), the two-way interaction effect of perceived behavioral plasticity and country (second row), the two-way interaction effect of perceived behavioral plasticity and income (third row), and the three-way interaction effect of perceived behavioral plasticity, income, and country (fourth row) across Cauchy prior scales.



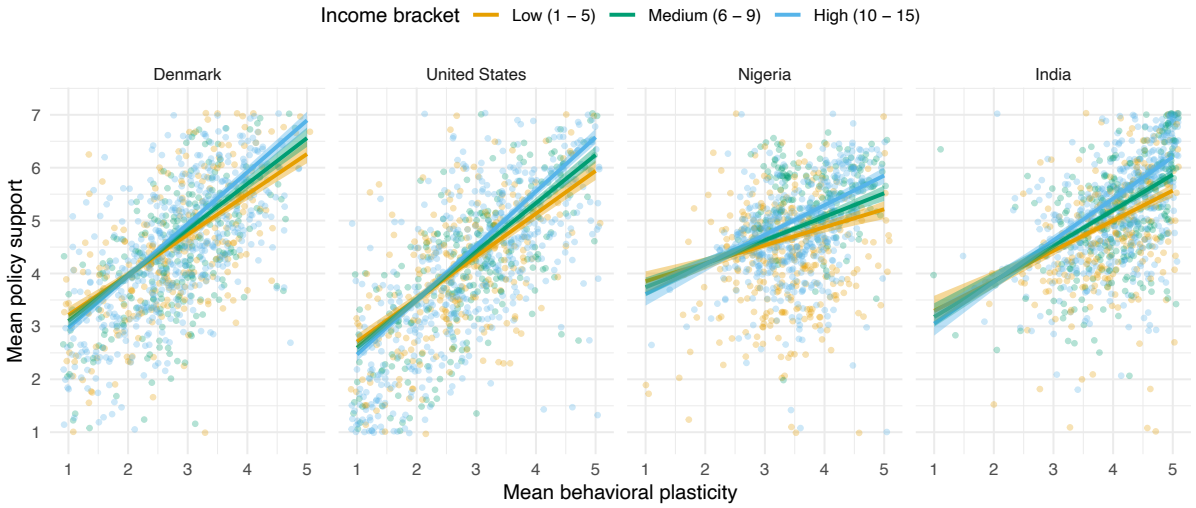
Supplementary Fig. 7. Posterior mean and 95% credible intervals of (loosely) domain-matched mean policy support across behaviors, income groups, and countries. Points indicate empirical means within income groups and levels of perceived behavioral plasticity. Using less energy for heating and cooling, driving fewer kilometers, and eating less red or white meat were loosely domain-matched, respectively, with the policies to increase energy efficiency requirements for buildings, expand public transport, and increase subsidies for food products with low greenhouse gas emissions.



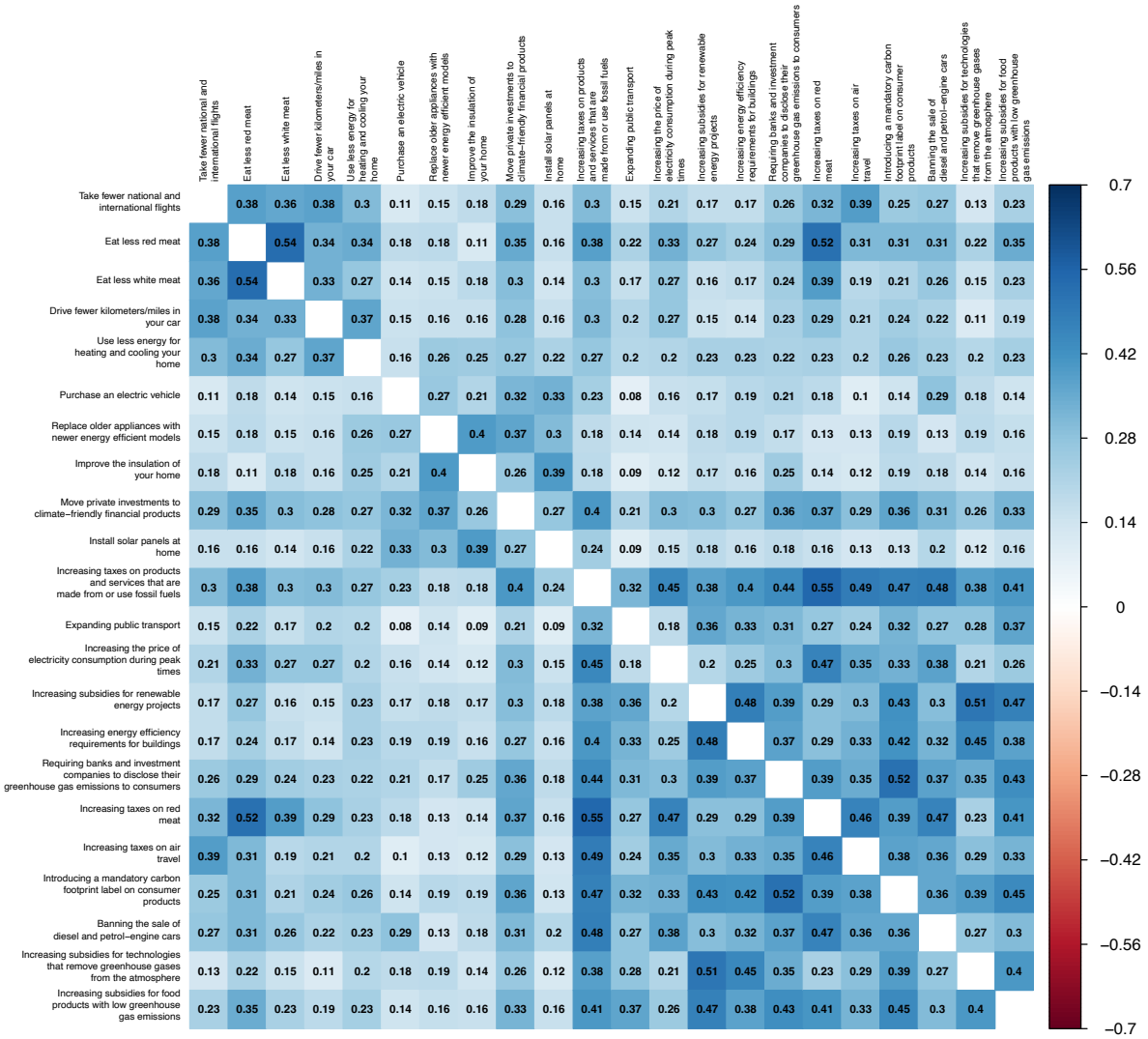
Supplementary Fig. 8. Posterior mean and 95% credible intervals of (loosely) domain-matched mean policy support across behavioral plasticity of actions, income groups, and countries. Points indicate empirical means within income groups and levels of behavioral plasticity. Improving the insulation of one's home, replacing older appliances with newer energy-efficient ones, installing solar panels at home, purchasing an electric vehicle, and moving private investments to climate-friendly financial products were loosely domain-matched, respectively, with the policies to increase energy efficiency requirements for buildings, increase the price of electricity consumption during peak times, increase subsidies for renewable energy projects, ban the sale of diesel and petrol-engine cars, and require banks and investment companies to disclose their greenhouse gas emissions to consumers.



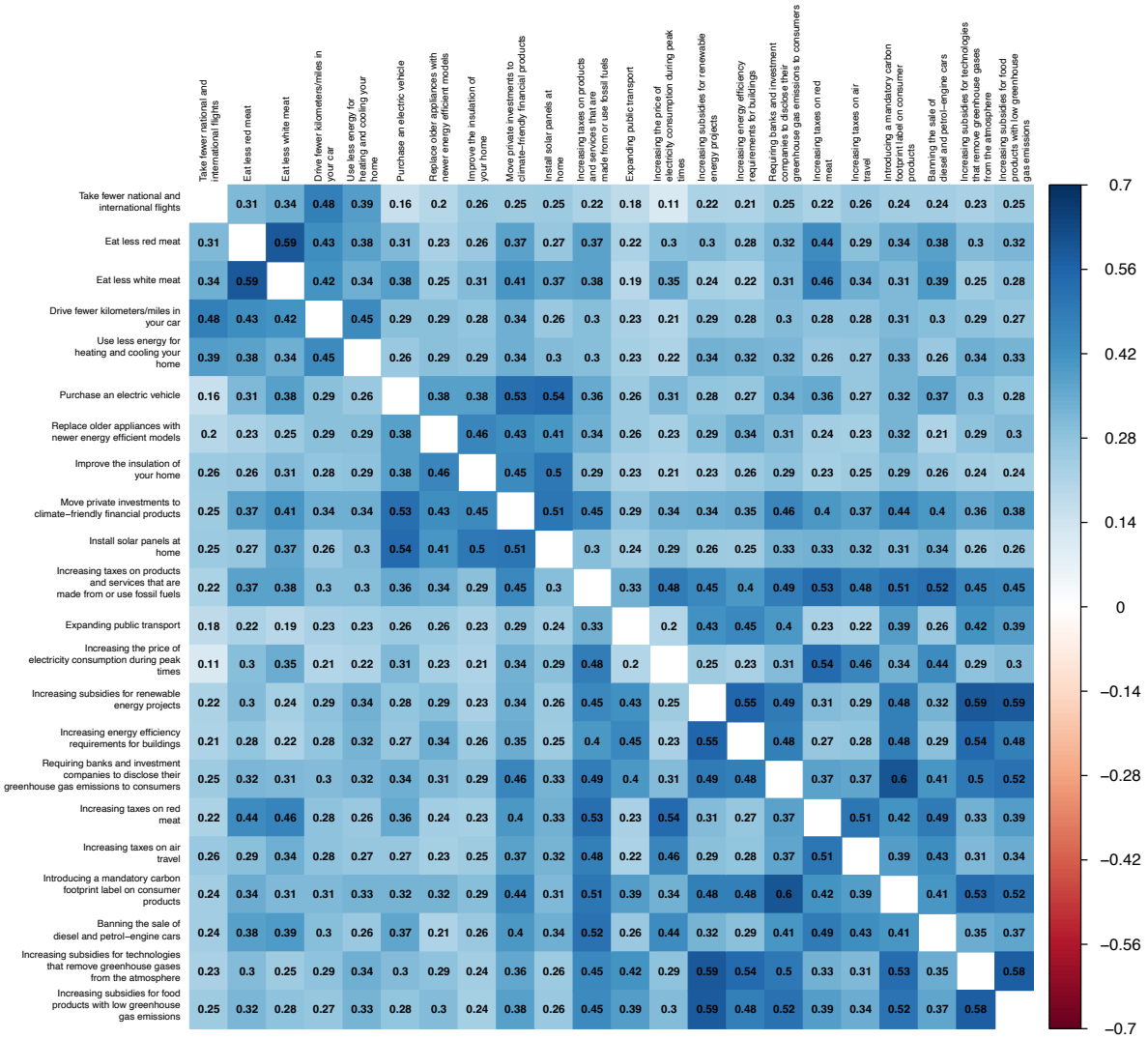
Supplementary Fig. 9. Relationship between mean behavioral plasticity and mean policy support across income levels. Posterior mean and 95% credible intervals of mean policy support across mean behavioral plasticity across income groups and countries. Points indicate empirical data.



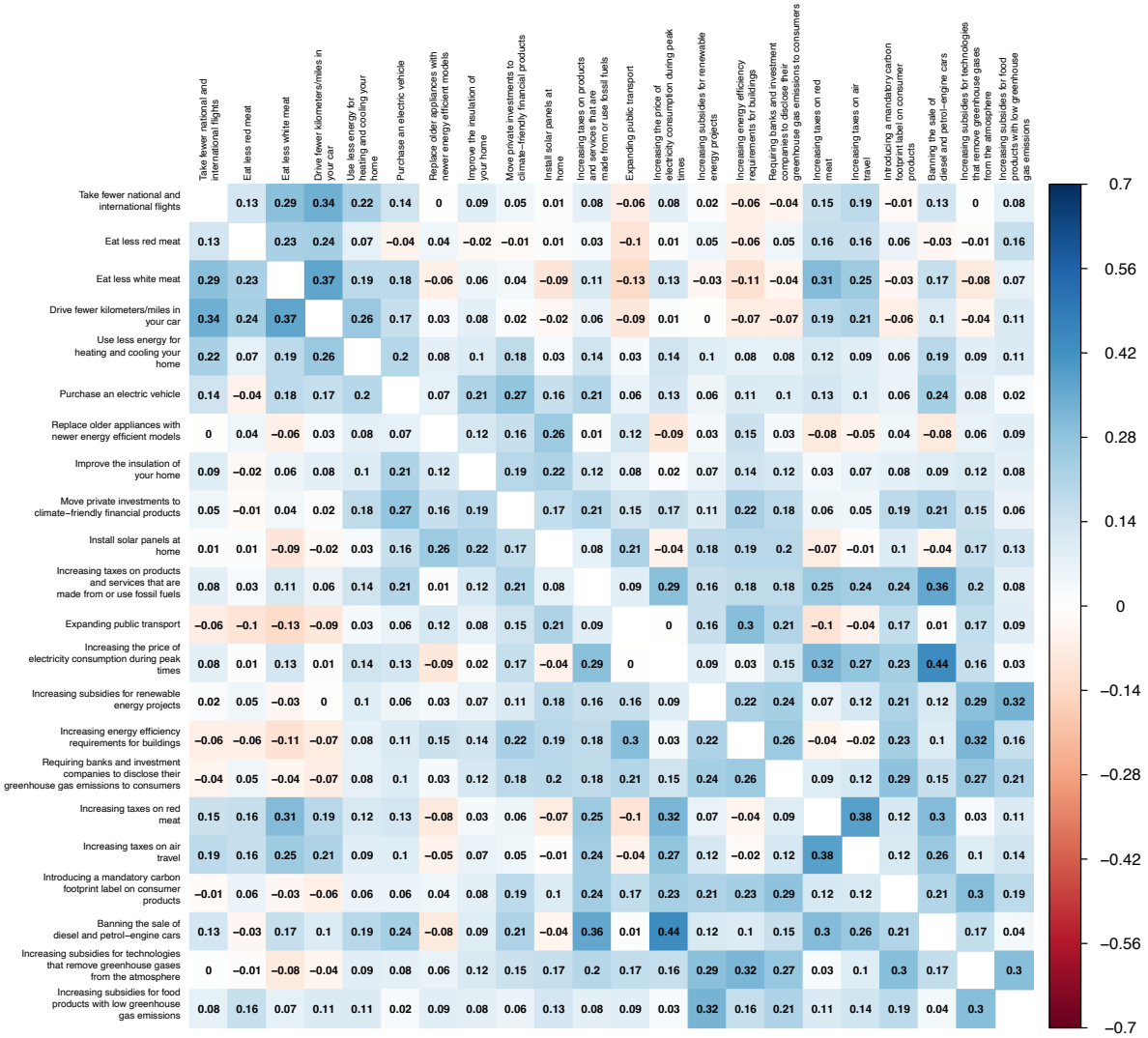
Supplementary Fig. 10. Kendall's correlations between behaviors and policy support for Denmark.



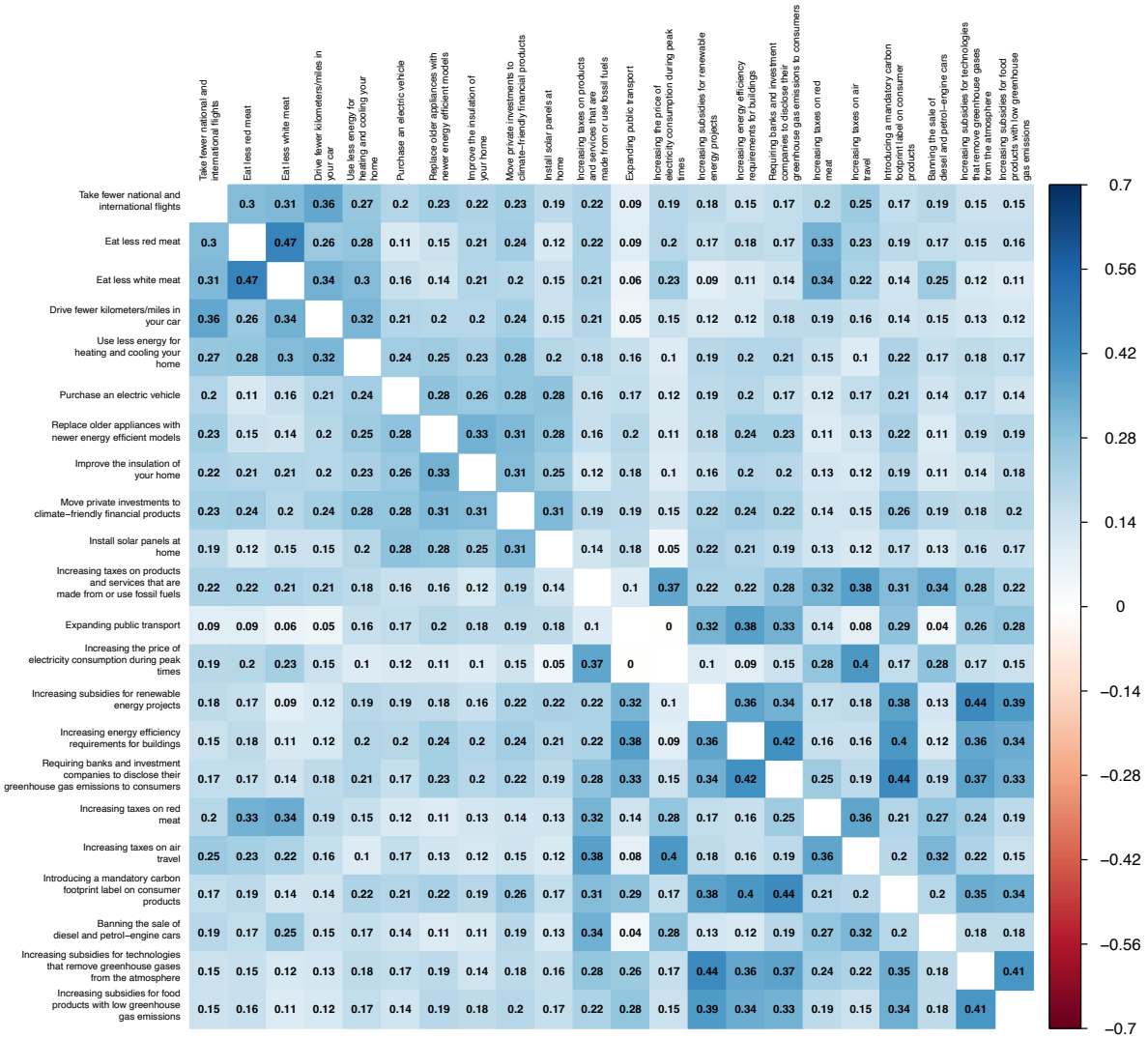
Supplementary Fig. 11. Kendall's correlations between behaviors and policy support for the United States.



Supplementary Fig. 12. Kendall's correlations between behaviors and policy support for Nigeria.



Supplementary Fig. 13. Kendall's correlations between behaviors and policy support for India.



Supplementary Table 1. Descriptive statistics. Statistics for continuous variables are reported with mean, standard deviation, and median, whereas categorical variables are reported as percentages.

	Denmark				India				Nigeria				United States			
	N	Mean	SD	Median	N	Mean	SD	Median	N	Mean	SD	Median	N	Mean	SD	Median
Socio-demographics																
Age	1,001	52.80	15.42	53	1,001	31.11	9.62	29	1,001	35.43	10.51	34	1,000	50.07	17.85	48
Income group	1,001				1,001				1,001				1,000			
… General population	501	50%			501	50%			500	50%			500	50%		
… Top 10% of income	500	50%			500	50%			501	50%			500	50%		
Income (15 categories)	1,001	8.27	3.33	9.00	1,001	9.01	4.34	9.00	1,001	8.00	4.66	10.00	1,000	8.32	3.66	9.50
Gender	1,001				1,001				1,001				1,000			
... Male	540	54%			587	59%			738	74%			408	41%		
... Female	460	46%			411	41%			261	26%			583	58%		
... Non-binary / third gender	1	0.1%			2	0.2%			1	0.1%			9	0.9%		
... Prefer not to say	0	0%			1	0.1%			1	0.1%			0	0%		
Education	1,001				1,001				1,001				1,000			
... No schooling completed	1	0.1%			1	0.1%			1	0.1%			3	0.3%		
... Primary school	3	0.3%			3	0.3%			1	0.1%			6	0.6%		
... Lower secondary school	51	5%			7	0.7%			9	0.9%			9	0.9%		

... Vocational degree	241	24%			20	2%			32	3%			59	6%		
... High school	89	9%			59	6%			149	15%			221	22%		
... College degree	303	30%			487	49%			599	60%			428	43%		
... Master's degree or above	313	31%			424	42%			210	21%			274	27%		
Political orientation (left-right)	993	4.17	1.51	4	1,001	4.91	1.55	5	1,001	4.95	1.60	5	998	4.25	1.75	4.00
Perceived behavioral plasticity																
Purchase an electric vehicle	780	2.62	1.42	3.00	857	4.24	1.10	5	941	3.39	1.44	4	812	2.63	1.56	3.00
Replace older appliances with newer energy-efficient models	729	3.44	1.28	4	853	4.21	1.07	5	804	4.09	1.09	4.00	762	3.36	1.35	3.00
Improve the insulation of your home	522	2.83	1.42	3.00	886	4.05	1.06	4.00	920	3.83	1.15	4.00	742	3.19	1.47	3.00
Move private investments to climate-friendly financial products	743	2.62	1.30	3	932	4.06	1.08	4.00	947	3.78	1.21	4	851	2.63	1.42	3
Install solar panels at home	534	2.36	1.41	2.00	841	4.33	1.07	5	693	4.28	1.04	5	738	2.72	1.56	3.00
Take fewer national or international flights	860	2.83	1.42	3.00	923	3.80	1.20	4	965	3.29	1.35	3	812	3.10	1.50	3.00
Eat less red meat	944	2.88	1.43	3.00	727	3.78	1.37	4	962	3.41	1.45	4.00	948	2.71	1.51	3.00
Eat less white meat	972	2.38	1.26	2.00	827	3.65	1.34	4	977	2.89	1.47	3	976	2.32	1.37	2.00

Drive fewer kilometers in your car	870	2.61	1.37	3.00	935	3.93	1.21	4	957	3.13	1.45	3	948	3.06	1.51	3.00
Use less energy for heating or cooling your home	985	3.20	1.34	3	950	4.21	0.99	5.00	968	3.55	1.27	4.00	992	3.30	1.36	3.00
Climate policy support																
Climate policy support (composite)	1,001	4.40	1.23	4.42	1,001	5.06	1.04	5.08	1,001	4.73	0.97	4.75	1,000	4.04	1.41	4.17
Carbon tax	1,001	4.23	1.78	4	1,001	4.60	1.92	5	1,001	4.48	1.84	5	1,000	3.60	2.07	4.00
Expand public transport	1,001	4.98	1.72	5	1,001	6.05	1.38	7	1,001	5.66	1.49	6	1,000	5.09	1.64	5.00
Increase price of peak electricity consumption	1,001	3.12	1.79	3	1,001	3.76	2.05	4	1,001	4.08	2.09	4	1,000	3.05	1.94	3.00
Subsize renewable energy	1,001	5.50	1.55	6	1,001	5.89	1.53	7	1,001	5.18	1.71	5	1,000	4.85	1.92	5.00
Strengthen energy efficiency requirements in buildings	1,001	5.20	1.45	5	1,001	5.74	1.40	6	1,001	5.50	1.48	6	1,000	5.04	1.70	5.00
Mandate GHG disclosure by banks and investment companies	1,001	4.21	1.80	4	1,001	5.46	1.56	6	1,001	5.14	1.65	5	1,000	4.31	1.96	4.00
Tax on red meat	1,001	3.25	2.00	3	1,001	4.64	2.10	5	1,001	3.82	1.92	4	1,000	2.88	1.96	2.00
Tax on air travel	1,001	4.30	1.95	4	1,001	4.19	1.97	4	1,001	4.13	1.87	4	1,000	3.26	1.98	3.00
Introduce mandatory carbon footprint label	1,001	4.46	1.71	4	1,001	5.48	1.55	6	1,001	5.04	1.62	5	1,000	4.19	1.94	4.00

Ban diesel and petrol cars	1,001	3.27	1.95	3	1,001	3.97	2.03	4	1,001	3.82	2.16	4	1,000	2.94	1.96	3.00
Subsidize CDR technologies	1,001	5.39	1.49	6	1,001	5.44	1.79	6	1,001	5.15	1.66	5	1,000	4.80	1.86	5.00
Subsidize low-impact foods	1,001	4.85	1.71	5	1,001	5.52	1.66	6	1,001	4.80	1.84	5	1,000	4.51	1.93	5.00